Genetics: From Genes to Genomes

Fifth Edition

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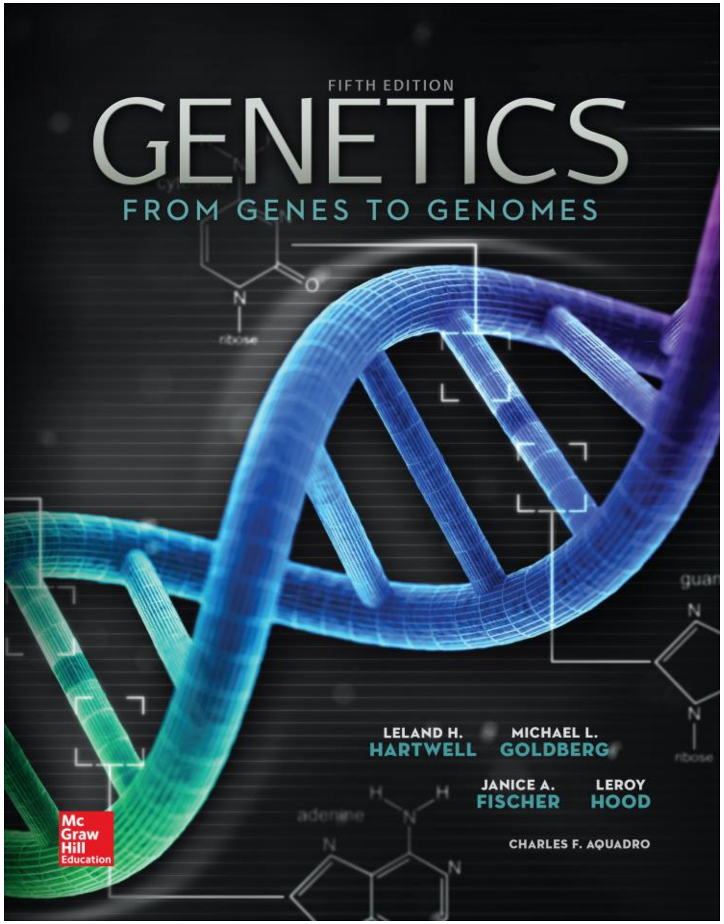
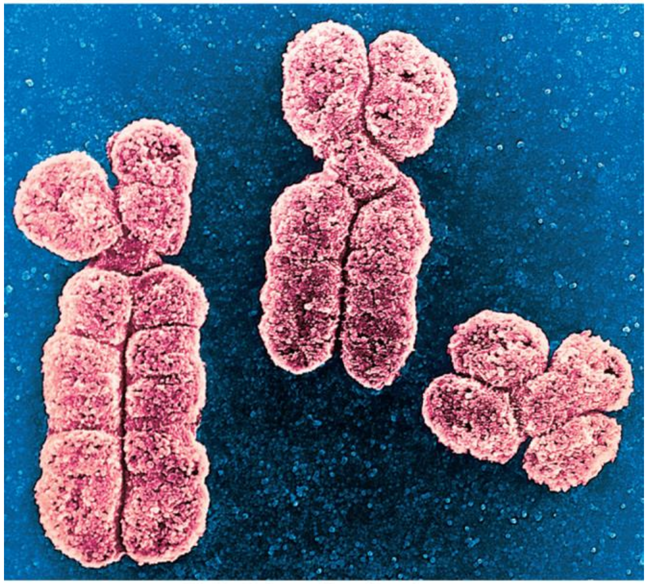


The Chromosome Theory of Inheritance



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CHAPTER OUTLINE

- 4.1 Chromosomes: The Carriers of Genes
- 4.2 Sex Chromosomes and Sex Determination
- 4.3 Mitosis: Cell Division That Preserves Chromosome Number
- 4.4 Meiosis: Cell Divisions That Halve Chromosome Number
- 4.5 Gametogenesis
- 4.6 Validation of the Chromosome Theory
- 4.7 Sex-Linked and Sexually Dimorphic Traits in Humans

Down syndrome is attributable to an abnormal number of chromosomes.

How can an extra chromosome cause cause a wide range of phenotypic effects?

- **Chromosomes transmit genetic information.**
- **The type and amount of genetic material is important for normal development.**

Trisomy 21 results in Down syndrome

4.1 Chromosomes: The carriers of genes

Learning Objectives:

- **Determine the number and origin of chromosomes in somatic cells, gametes, and zygotes.**
- **Distinguish between homologous and nonhomologous chromosomes.**
- **Distinguish between sister chromatids and nonsister chromatids.**

Evidence that genes reside in the nucleus

1667 - Anton Van Leeuwenhoek

- **Microscopy revealed that semen contain spermatozoa ("sperm animals")**
- **Hypothesized that sperm may enter egg to achieve fertilization**

1854 – 1874

- **Direct observations of fertilization through union of nuclei of eggs and sperm (frog and sea urchin)**
- **Conclusion: something in the nucleus must contain the hereditary material**

Evidence that genes reside in chromosomes

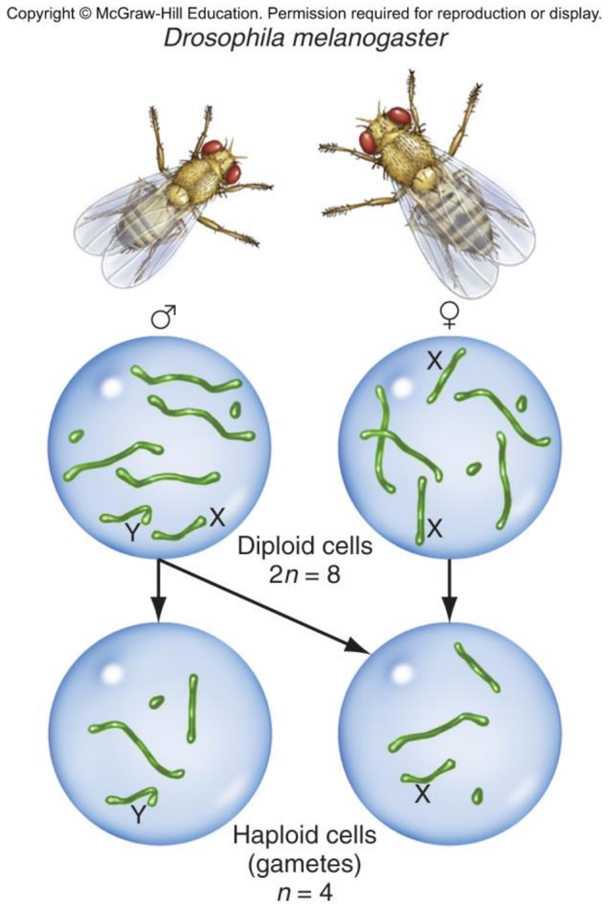
1880s – improved microscopy and staining techniques

- **Long, threadlike bodies (chromosomes) visualized in the nucleus**
- **Movement of these bodies followed through cell division**

Mitosis - nuclear division that generates two daughter cells containing the same number and type of chromosomes as parent cell

Meiosis - Nuclear division that generates gametes (egg and sperm) containing half the number of chromosomes found in other cells

Diploid versus haploid: *2n* versus *n*



Most body cells are diploid (2n) (each chromosome pair has one maternal and one paternal copy)

Meiosis → haploid (*n*) gametes

In Drosophila, *2n* = 8, *n* = 4

In humans , *2n* = 46 and *n* = 23

Hartwell et al., 5th ed., Chapter 4

Fertilization is the union of haploid gametes to produce diploid zygotes

Meiosis generates haploid gametes (*n*) – carry only a single set of chromosomes

Two gametes fuse during fertilization to form a zygote.

Zygotes are diploid (*2n*) – carry two matching sets of chromosomes

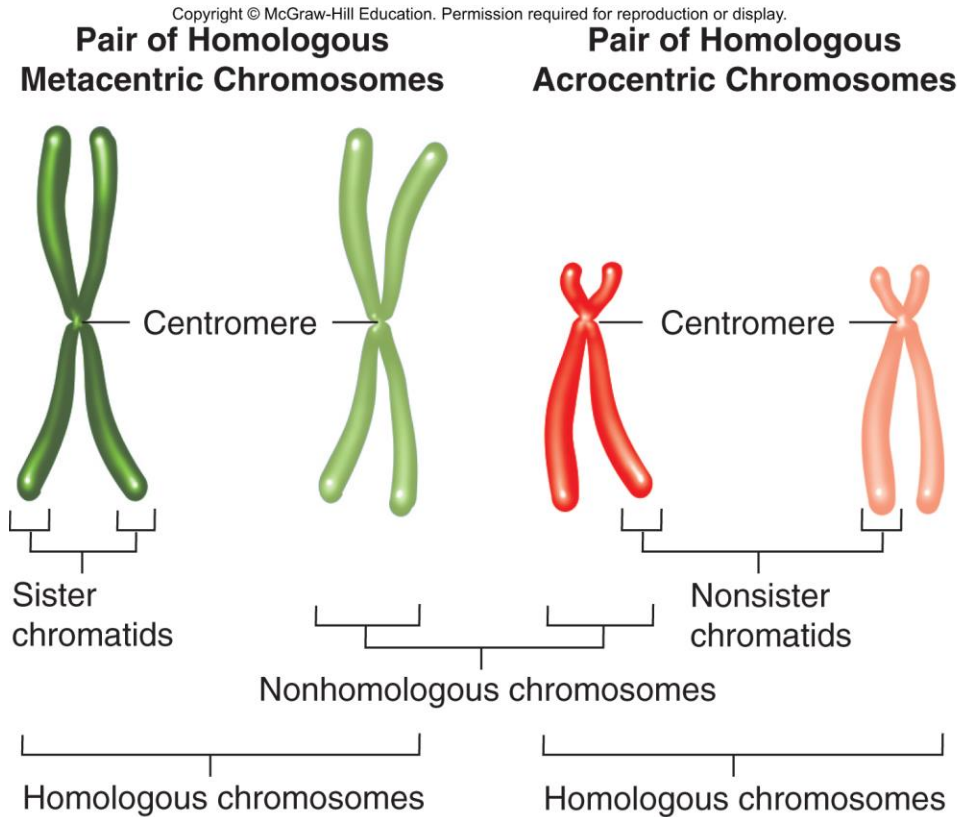
Metaphase chromosomes can be classified by centromere position

Metacentric chromosome

- **centromere is in the middle**

Acrocentric chromosome

- **centromere is near one end**



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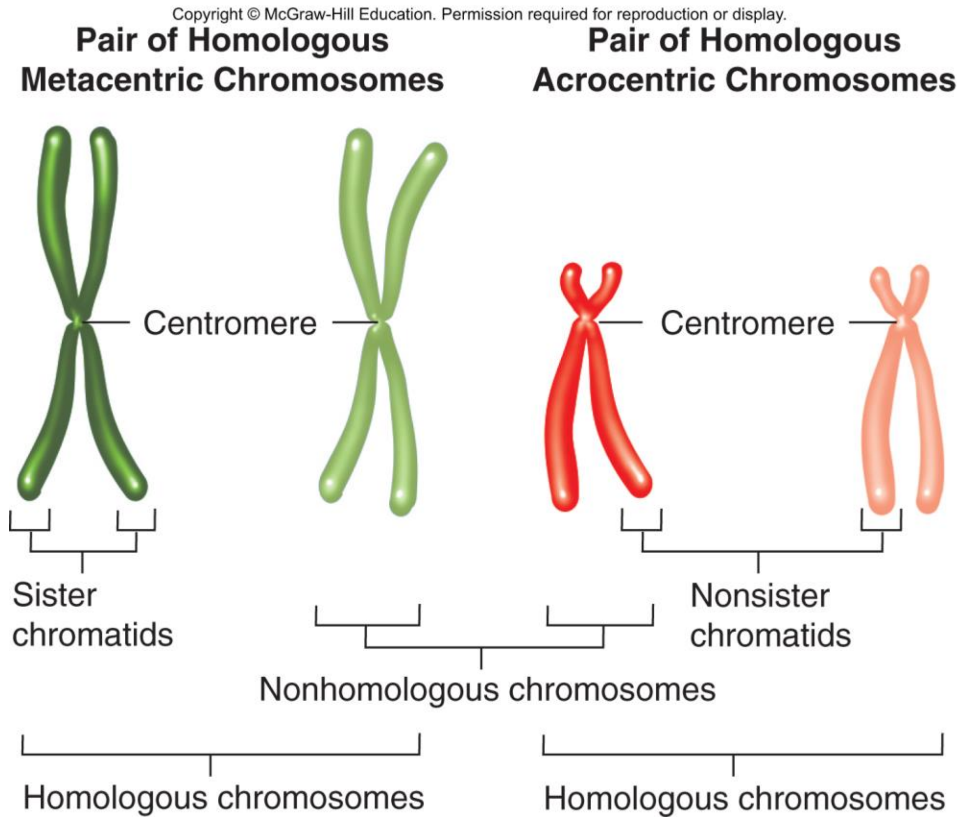
Metaphase chromosomes can be classified by centromere position

Sister chromatids are identical copies of a replicated chromosome

Homologs contain the same set of genes, but can have different alleles for some genes

Nonhomologs carry completely unrelated sets of genes



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Karyotype: Micrograph of strained chromosomes arranged in homologous pairs

Homologous chromosomes—same size, shape, and banding

Each species has a characteristic diploid number of chromosomes (fruit flies 2n=8; dogs 2n=78)

Autosomes- all chromosomes except X and Y

Sex chromosomes- unpaired X and Y

Karyotype of a human male ($2n = 46$, $n = 23$)

4.2 Sex chromosomes and sex determination

Learning Objectives:

- **Predict the sex of humans with different complements of X and Y chromosomes.**
- **Describe the basis of sex reversal in humans.**
- **Compare the means of sex determination in different organisms.**

One chromosome pair determines sex in grasshoppers

W. S. Sutton studied meiosis in great lubber grasshoppers

Before meiosis, testes cells had 24 chromosomes

- 22 in matched pairs (autosomes) and 2 unmatched (large = X and smaller = Y)

After meiosis, two types of sperm were formed

- 1/2 of sperm had 11 chromosomes and an X
- 1/2 of sperm had 11 chromosomes and a Y

After meiosis, only one type of egg was produced

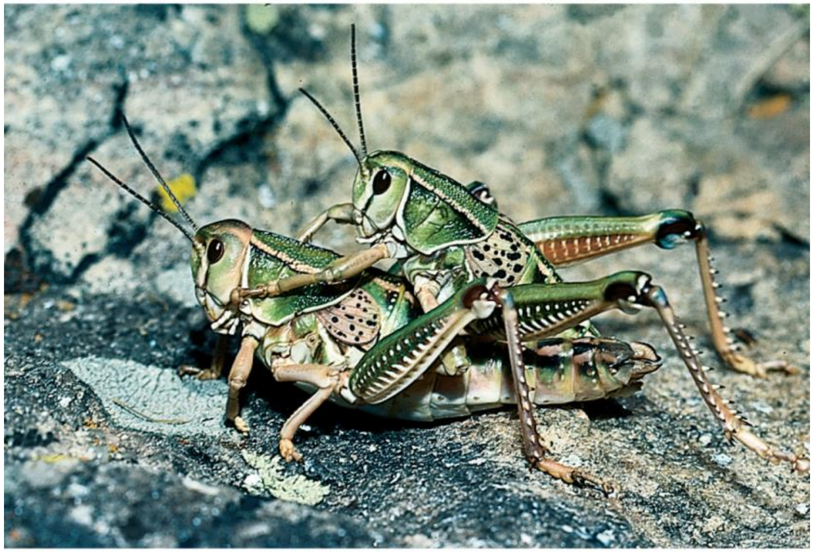
- All had 11 chromosomes plus an X

The great lubber grasshopper

Fertilization of egg with sperm carrying an X → XX female

Fertilization of egg with sperm carrying a Y → XY male

Sutton concluded that the X and Y chromosomes determine sex



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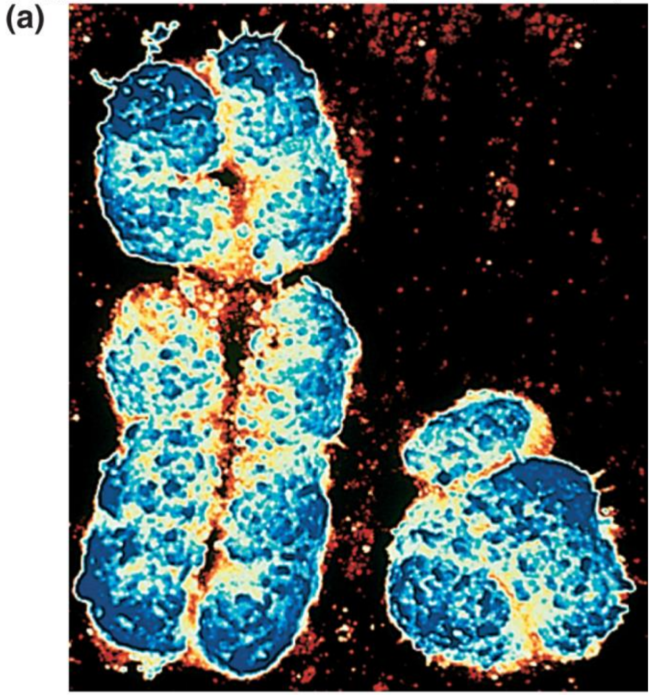
The X and Y chromosomes determine sex in humans

Children receive an X chromosome from their mother, but either an X or Y chromosome from their father

Results in 1:1 ratio of females-to-males



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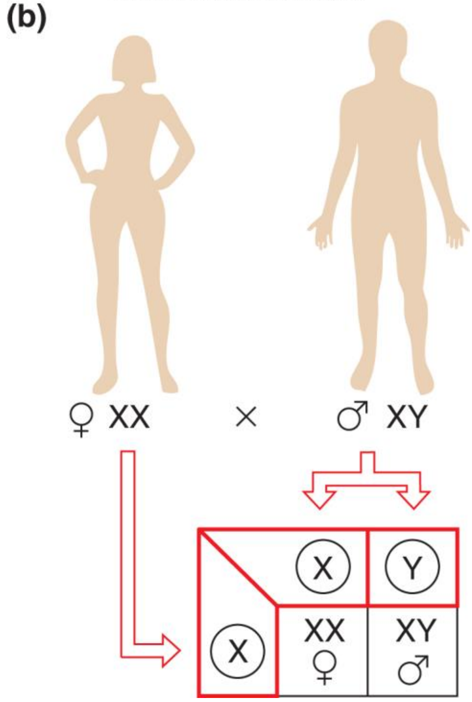
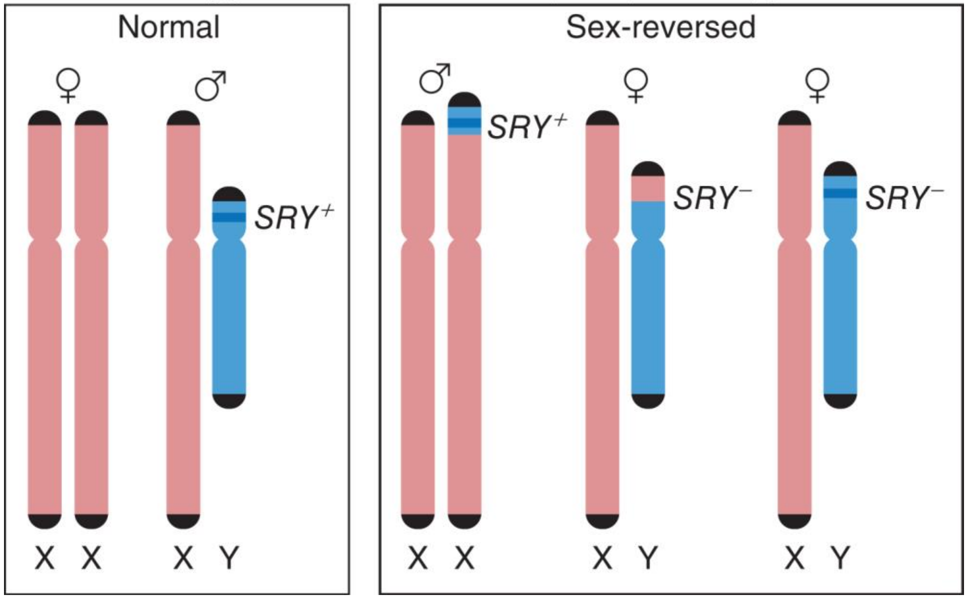


A small portion of the Y chromosome determines maleness in humans

The primary determinant of maleness is a single gene called *SRY* (*sex determining region of Y*)

Sex reversal provided evidence implicating *SRY*

- ***SRY* is always present in XX males**
- ***SRY* is always nonfunctional in XY females**



Sex determination in fruit flies and humans

In *Drosophila*, the ratio of X chromosomes to autosomes determines gender

In humans, presence or absence of Y chromosome determines gender

Abnormal numbers of X or Y chromosomes have different effects in humans and flies

TABLE 4.1 Sex Determination in Fruit Flies and Humans

	Complement of Sex Chromosomes						
	XXX	**XX**	**XXY**	**XO**	**XY**	**XYY**	**OY**
Drosophila	Dies	Normal female	Normal female	Sterile male	Normal male	Normal male	Dies
Humans	Nearly normal female	Normal female	Klinefelter male (sterile); tall, thin	Turner female (sterile); webbed neck	Normal male	Normal or nearly normal male	Dies

Humans can tolerate extra X chromosomes (e.g., XXX) better than can *Drosophila* because in humans all but one X chromosome becomes a Barr body, as discussed later in this chapter. Complete absence of an X chromosome is lethal to both fruit flies and humans. Additional Y chromosomes have little effect in either species. Although the Y chromosome in *Drosophila* does not determine whether a fly looks like a male, it is necessary for male fertility; XO flies are thus sterile males.

Mechanisms of sex determination differ between species

Heterogametic sex – gender with two different kinds of gametes (e.g. XY males in humans, ZW females in birds)

Homogametic sex – gender with one type of gamete (e.g. XX females in humans, ZZ males in birds)

In some species, gender is determined by environment (e.g. temperature)

TABLE 4.2 Mechanisms of Sex Determination

	♀	♂
Humans and *Drosophila*	XX	XY
Moths and *C. elegans*	XX (hermaphrodites in *C. elegans*)	XO
Birds and Butterflies	ZW	ZZ
Bees and Wasps	Diploid	Haploid
Lizards and Alligators	Cool temperature	Warm temperature
Tortoises and Turtles	Warm temperature	Cool temperature
Anemone Fish	Older adults	Young adults

In the species highlighted in *purple*, sex is determined by sex chromosomes. The species highlighted in *blue* have identical chromosomes in the two sexes, and sex is determined instead by environmental or other factors. Anemone fish (*bottom row*) undergo a sex change from male to female as they age.

4.3 Mitosis: Cell division that preserves chromosome number

Learning Objectives:

- **Describe the key chromosome behaviors during mitosis.**
- **Diagram the forces and structures that dictate chromosomal movement during mitosis.**

The cell cycle is a repeating pattern of cell growth and division

Nuclear division during mitosis apportions chromosomes in equal fashion to two genetically identical daughter cells

Interphase has three parts - gap 1 (G_1) phase, synthesis (S) phase, and gap 2 (G_2) phase

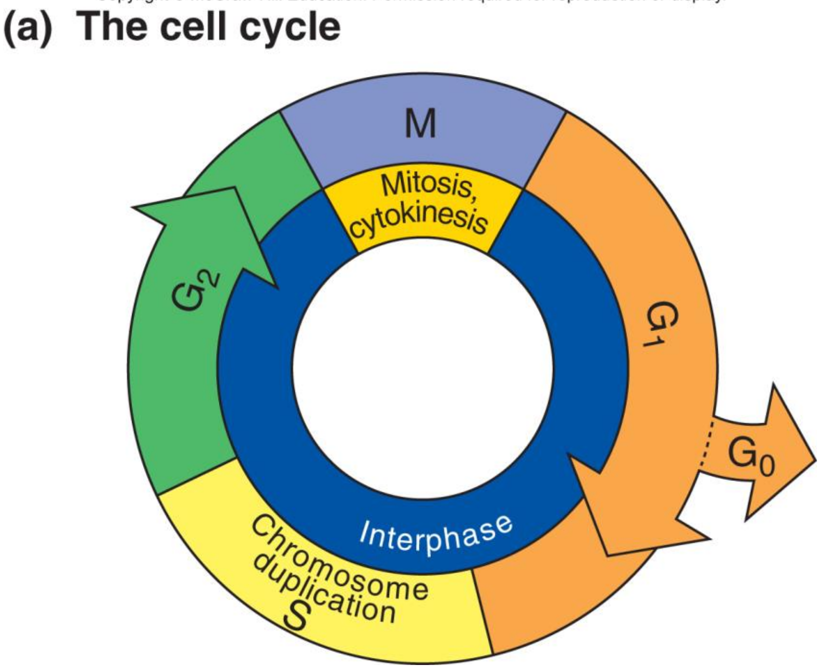
- **Period of cell growth and chromosome duplication between divisions**
- **Formation of microtubules in cytoplasm**
  - **Centrosome – microtubule organizing center near the nuclear envelope**
  - **Centrioles – core of centrosome, not found in plant cells**

The cell cycle: An alternation between interphase and mitosis

Most of cell growth occurs during G_1 and G_2 phases.

Some terminally differentiated cells stop dividing and arrest in G_0 stage.

Chromosomes replicate to form sister chromatids during S phase.



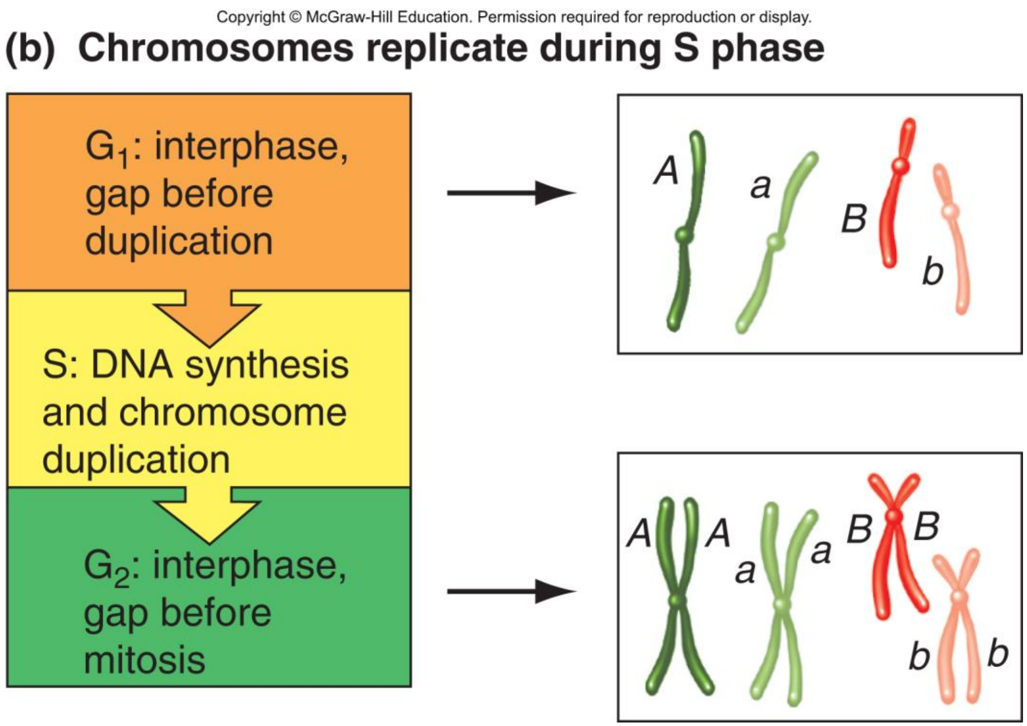
Chromosomes replicate during S phase

G_1 phase – chromosomes are not duplicating or dividing

- **Length of time varies in different cell types**

S phase – duplication of chromosome into sister chromatids

G_2 phase – synthesis of proteins required for mitosis



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Mitosis has five stages that have distinct cytological characteristics

The five stages of mitosis and their major events

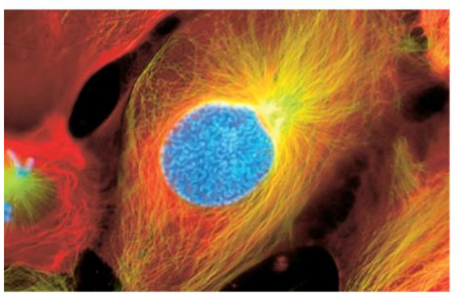
- **Prophase** - chromosomes condense and become visible
- **Prometaphase** – spindle forms and sister chromatids attach to microtubules from opposite centrosomes
- **Metaphase** – chromosome align at the cell equator
- **Anaphase** – sister chromatids separate and move to opposite poles
- **Telophase** – chromosomes decondense and are enclosed in two nuclei

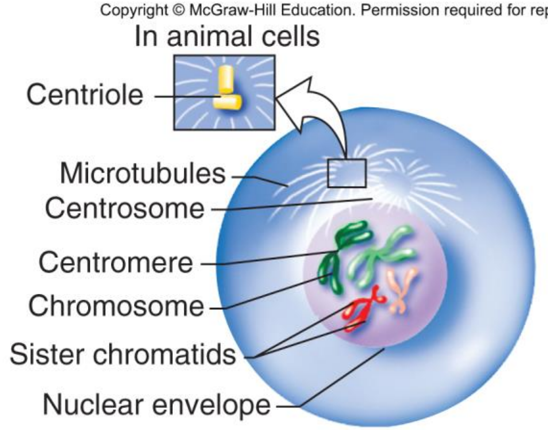
Stages of mitosis: prophase

Chromosomes condense and become visible

Centrosomes move apart toward opposite poles

Nucleoli begin to disappear





(a) Prophase: (1) Chromosomes condense and become visible; (2) centrosomes move apart toward opposite poles and generate new microtubules; (3) nucleoli begin to disappear.

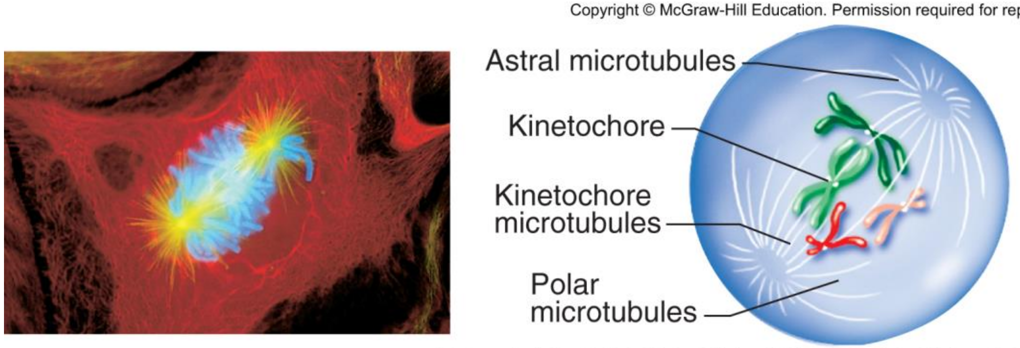
Photographs by Dr. Conly L. Rieder, Division of Molecular Medicine, Wadsworth Center, NYS Dept. Of Health, Albany, NY

Stages of mitosis: prometaphase

Nuclear envelope breaks down

Microtubules from centrosomes invade the nucleus and connect to kinetochores in centromere of each chromatid

- **Sister chromatids attach to microtubules from opposite poles**
- **Mitotic spindle forms from three kinds of microtubules (astral, kinetochore, and polar)**



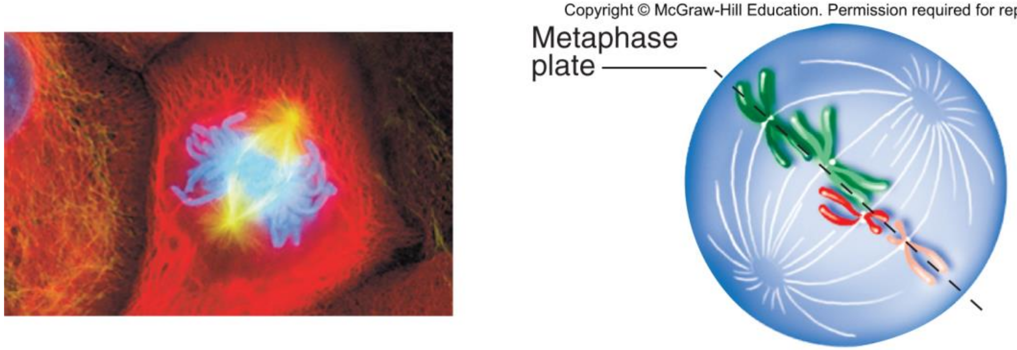
(b) Prometaphase: (1) Nuclear envelope breaks down; (2) microtubules from the centrosomes invade the nucleus; (3) sister chromatids attach to microtubules from opposite centrosomes.

Photographs by Dr. Conly L. Rieder, Division of Molecular Medicine, Wadsworth Center, NYS Dept. Of Health, Albany, NY

Stages of mitosis: metaphase

Chromosomes align on the metaphase plate with sister chromatids facing opposite poles

Forces pushing and pulling chromosomes to or from each pole are in balanced equilibrium



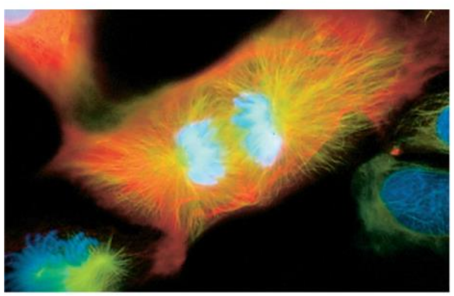
(c) Metaphase: Chromosomes align on the metaphase plate with sister chromatids facing opposite poles.

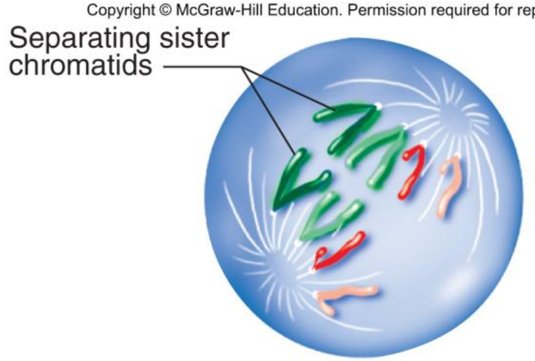
Photographs by Dr. Conly L. Rieder, Division of Molecular Medicine, Wadsworth Center, NYS Dept. Of Health, Albany, NY

Stages of mitosis: anaphase

Centromeres of all chromosomes divide simultaneously

Kinetochore microtubules shorten and pull separated sister chromatids to opposite poles (characteristic V shape)





(d) Anaphase: (1) Centromeres divide; (2) the now separated sister chromatids move to opposite poles.

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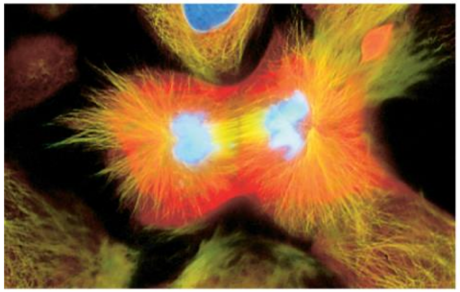
Stages of mitosis: telophase

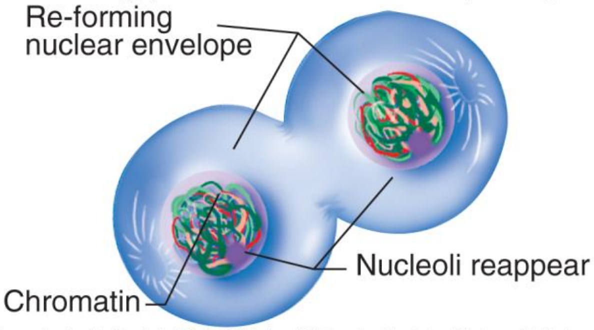
Nuclear envelope forms around each group of chromatids

Nucleoli re-form

Spindle fibers disperse

Chromosomes decondense





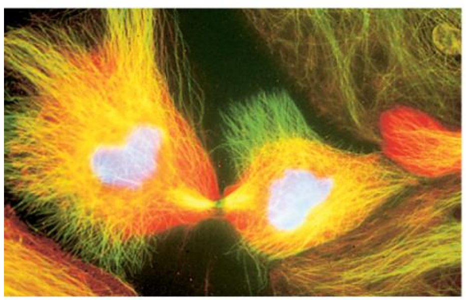
(e) Telophase: (1) Nuclear membranes and nucleoli re-form; (2) spindle fibers disappear; (3) chromosomes uncoil and become a tangle of chromatin.

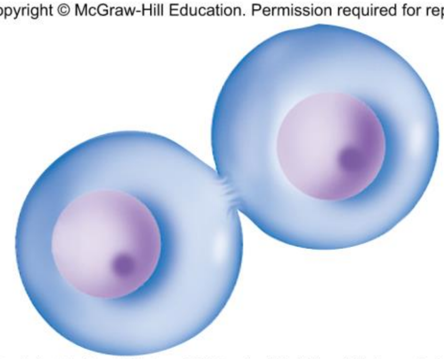
Photographs by Dr. Conly L. Rieder, Division of Molecular Medicine, Wadsworth Center, NYS Dept. Of Health, Albany, NY

Cytokinesis is the final stage of cell division

Begins during anaphase but not completed until after telophase

Parent cells split into two daughter cells with identical nuclei





(f) Cytokinesis: The cytoplasm divides, splitting the elongated parent cell into two daughter cells with identical nuclei.

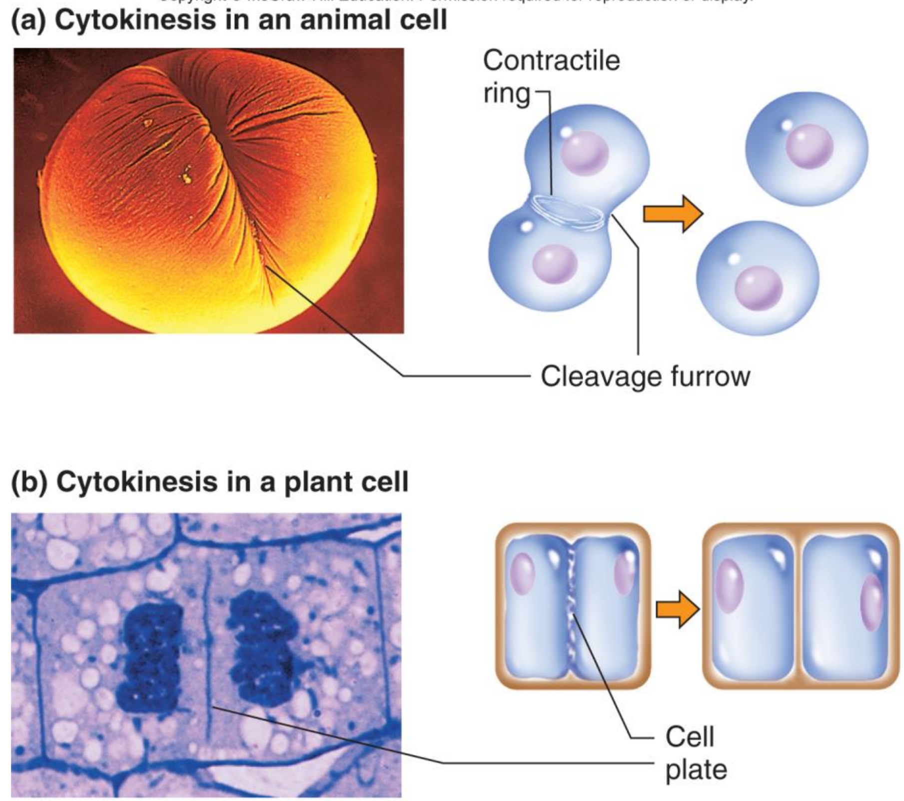
Photographs by Dr. Conly L. Rieder, Division of Molecular Medicine, Wadsworth Center, NYS Dept. Of Health, Albany, NY

Cytokinesis: The cytoplasm divides and produces two daughter cells

Animals have contractile ring that contracts to form cleavage furrow

Plants have cell plate that forms near equator of cell

Organelles (e.g. ribosomes, mitochondria, Golgi bodies) are distributed to each daughter cell

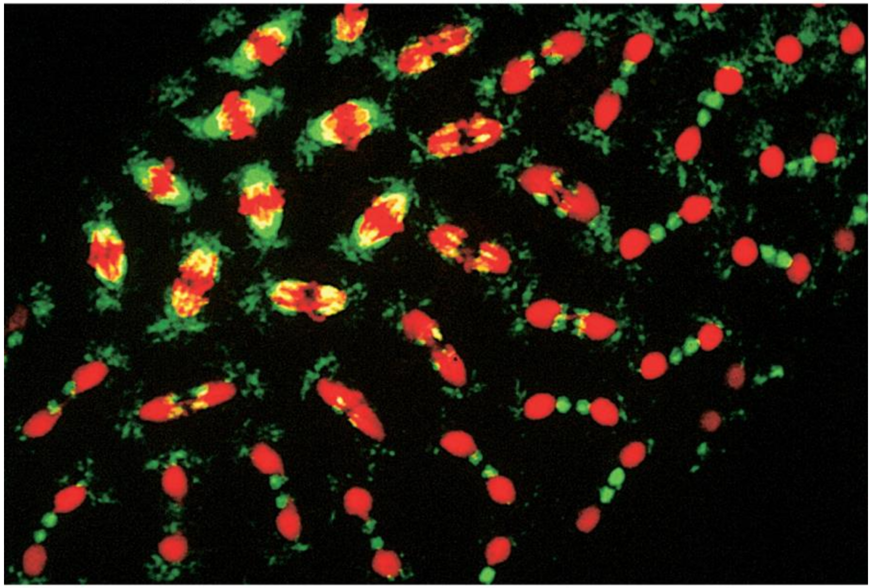


(a): © David M. Phillip/Visuals Unlimited; (b): © R. Calentine/Visuals Unlimited

Cytokinesis does not always occur after mitosis

In fertilized *Drosophila* eggs, 13 rounds of mitosis occur without cytokinesis

Results in syncytial embryo with thousands of nuclei within a single cell



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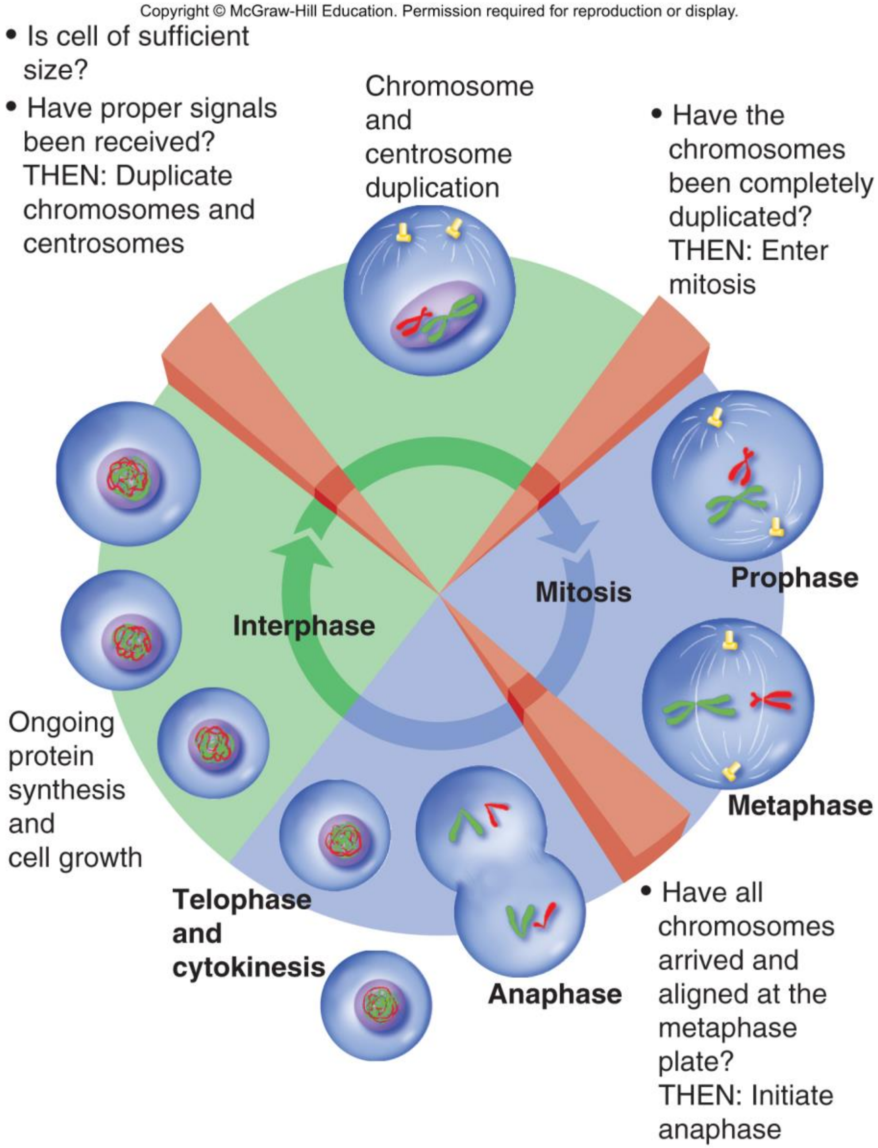
Checkpoints help regulate the cell cycle

At each checkpoint, prior events must be completed before the next step of the cycle can begin



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4.4 Meiosis: Cell division that halve chromosome number

Learning Objectives:

- **Describe the key chromosome behaviors during meiosis that lead to haploid gametes.**
- **Compare chromosome behaviors during mitosis and meiosis.**
- **Explain how chromosome alignment and crossing over contribute to the genetic diversity of gametes.**

Two general types of cells in plants and animals

Somatic cells make up vast majority of cells in the organism

- **In G_0 or are actively going through mitosis**

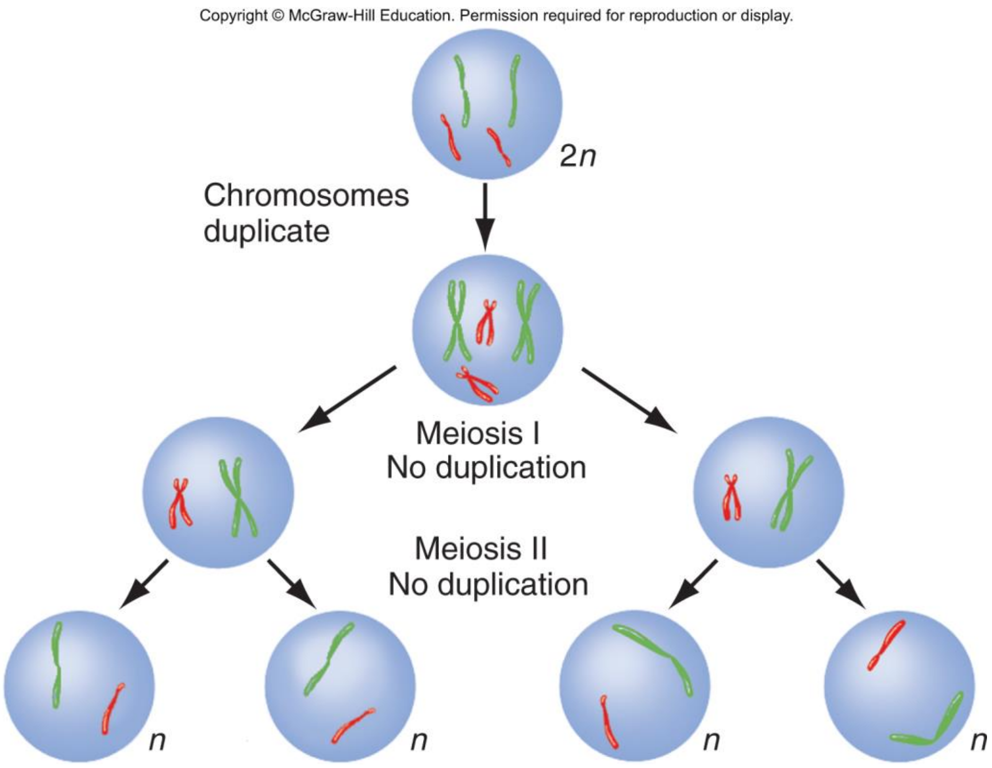
Germ cells are precursors to gametes

- **Set aside during embryogenesis**
- **Become incorporated into reproductive organs**
- **Only cells that undergo meiosis produce haploid gametes**

Overview of meiosis

Two rounds of cell division

- Chromosomes duplicate once; nuclei divide twice
- Meiosis I reduces the chromosomes from 2n to n.



Overview of meiosis I

Homologs pair, cross over, and then segregate

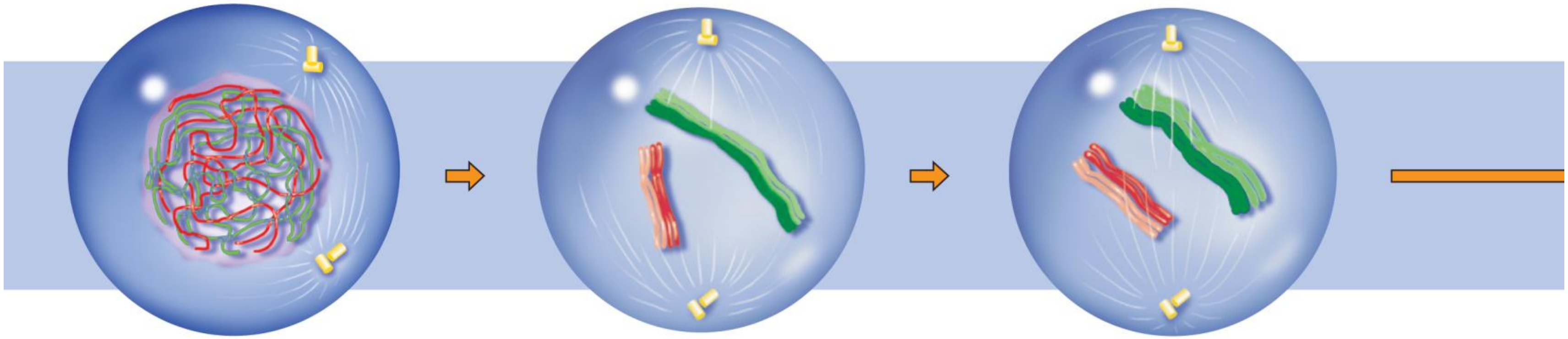
- Sister chromatids remain intact throughout meiosis I
- Maternal and paternal homologs recombine and create new combinations of alleles
- After recombination, homologs segregate to different daughter cells

The first three substages of prophase I: Leptotene, zygotene, and pachytene

Homologs pair and are held together by synaptonemal complex

Crossing-over (recombination) occurs during prophase I

Meiosis I: A reductional division



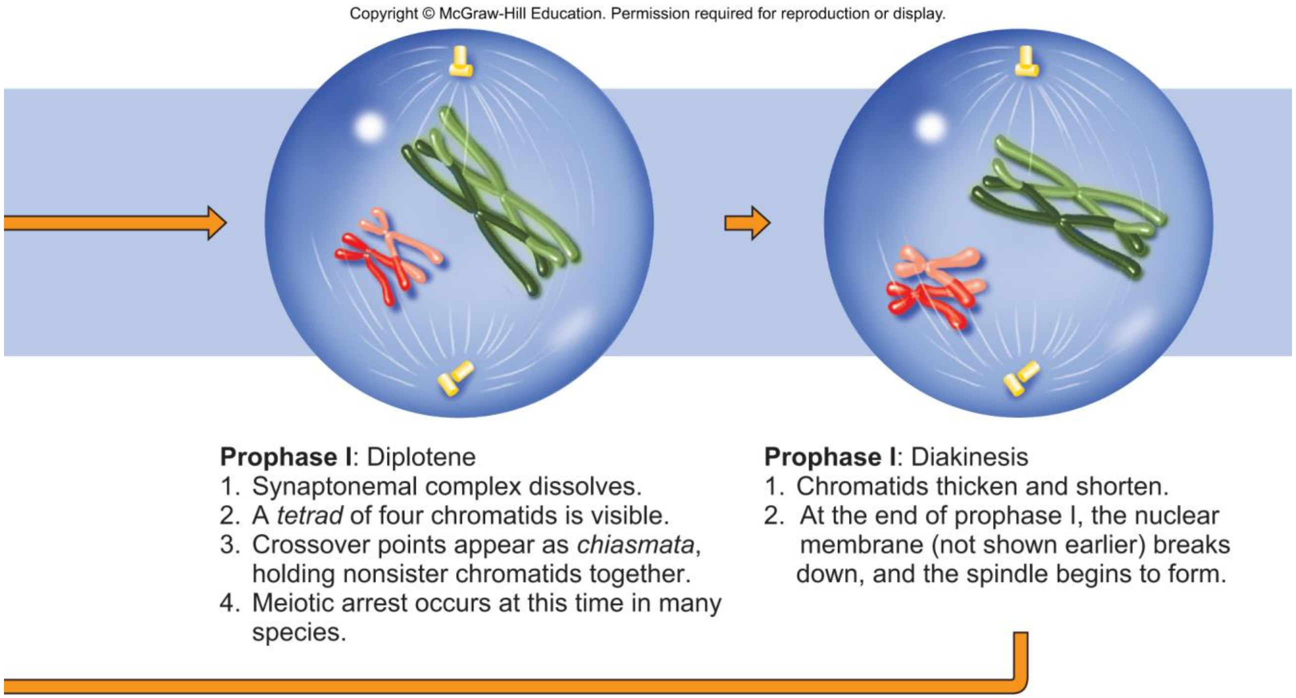
Prophase I: Leptotene
1. Chromosomes thicken and become visible, but the chromatids remain invisible.
2. Centrosomes begin to move toward opposite poles.

Prophase I: Zygotene
1. Homologous chromosomes enter *synapsis*.
2. The *synaptonemal complex* forms.

Prophase I: Pachytene
1. Synapsis is complete.
2. *Crossing-over*, genetic exchange between nonsister chromatids of a homologous pair, occurs.

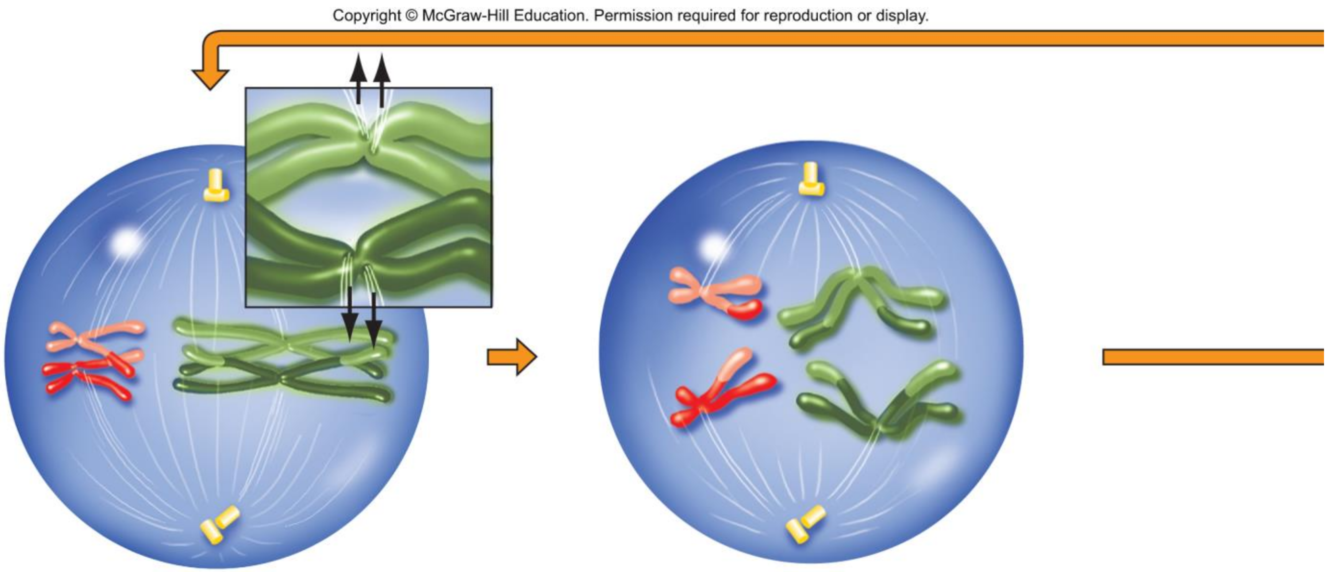
The last two substages of prophase : Diplotene and diakinesis

Synaptonemal complex dissolves and chromatids in each tetrad become visible



In metaphase I and anaphase I, homologs move to opposite poles

Note that the centromeres do not divide and sister chromatids are not separated



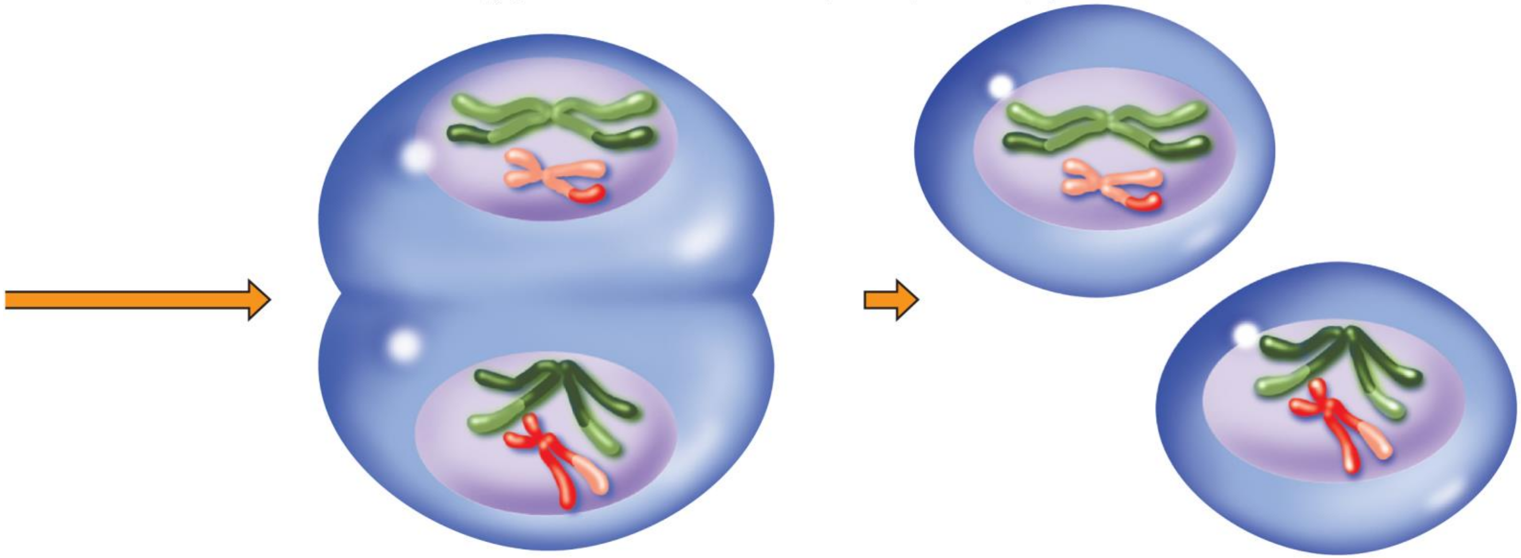
Metaphase I
1. Tetrads line up along the *metaphase plate*.
2. Each chromosome of a homologous pair attaches to fibers from opposite poles.
3. Sister chromatids attach to fibers from the same pole.

Anaphase I
1. The centromere does not divide.
2. The chiasmata dissolve.
3. Homologous chromosomes move to opposite poles.

Meiosis I is a reductional division

This cell had four chromosomes (2 pairs of homologs).

Now it has two (one copy of each nonhomolog).



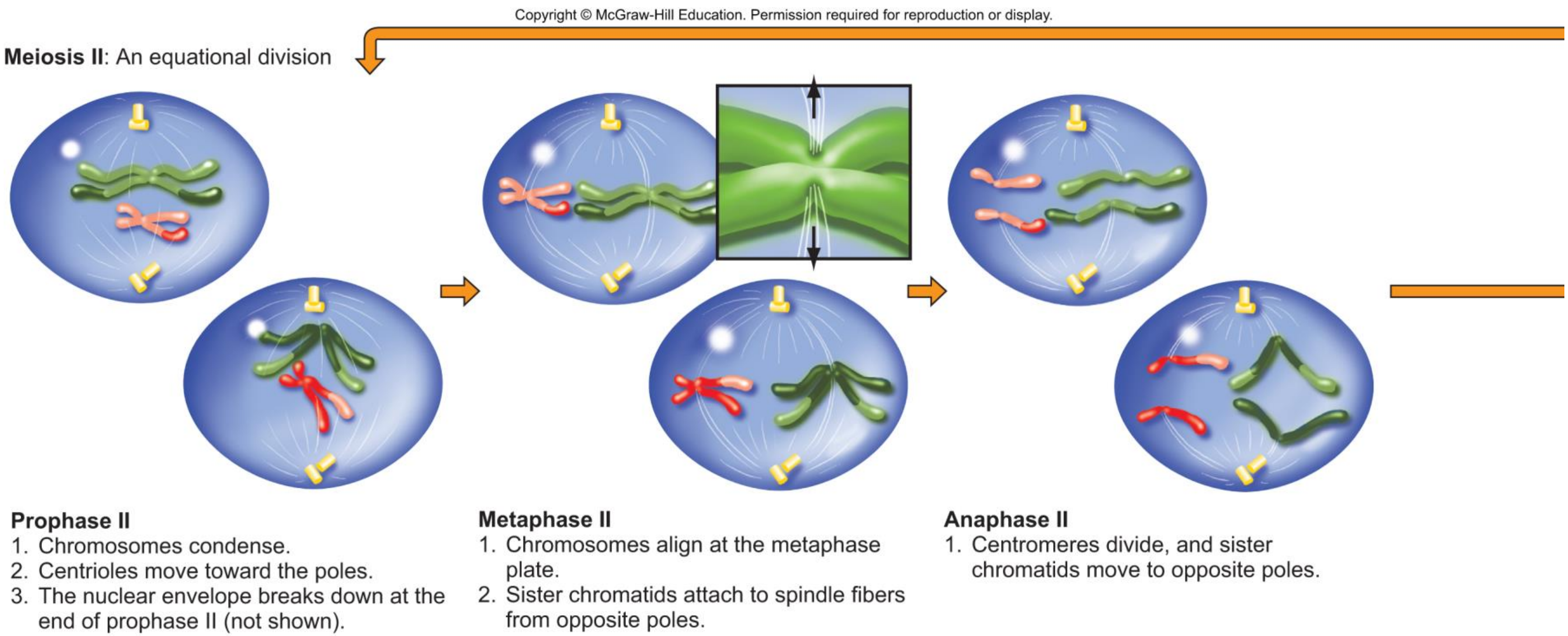
Telophase I

1. The nuclear envelope re-forms.
2. Resultant cells have half the number of chromosomes, each consisting of two sister chromatids.

Interkinesis

1. This is similar to interphase with one important exception: *No chromosomal duplication takes place*.
2. In some species, the chromosomes decondense; in others, they do not.

During meiosis II, sister chromatids separate and move to opposite poles



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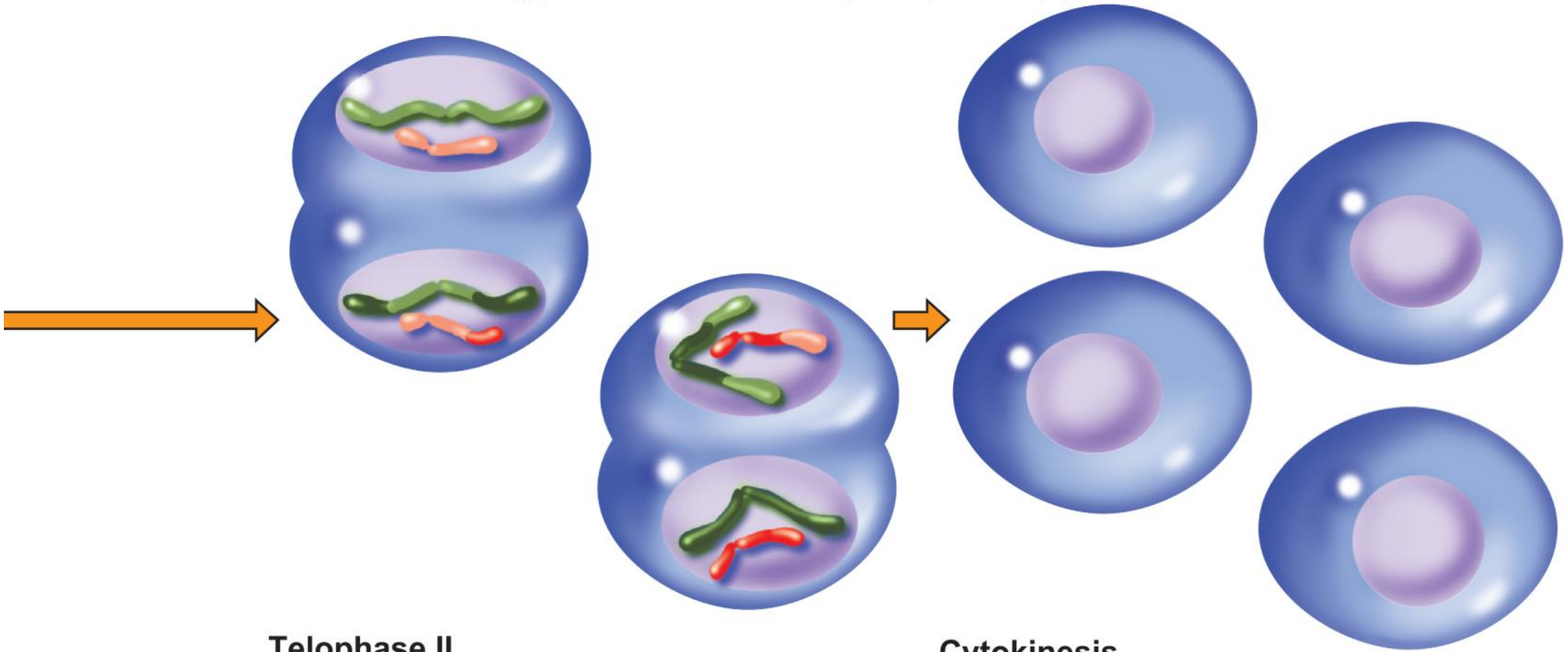
Meiosis II is an equational division



Telophase II

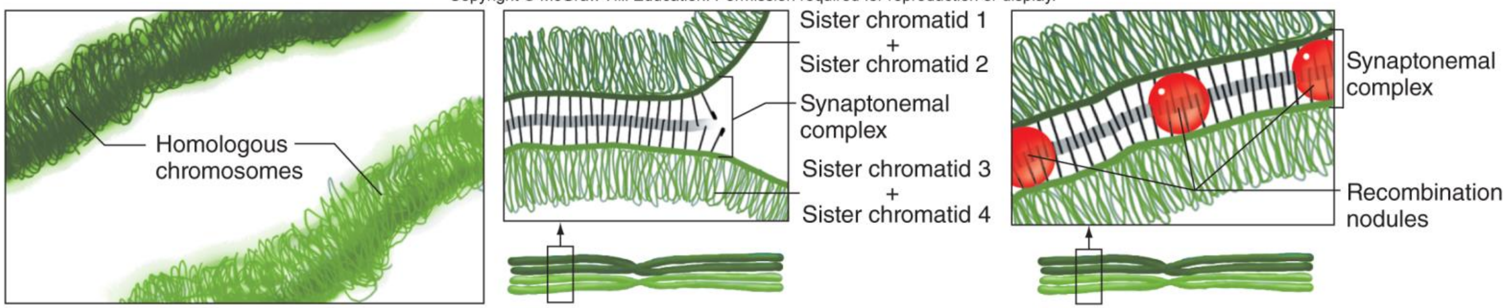
1. Chromosomes begin to uncoil.
2. Nuclear envelopes and nucleoli (not shown) re-form.

Cytokinesis

1. The cytoplasm divides, forming four new haploid cells.

Homolog pairing during meiosis I forms a tetrad

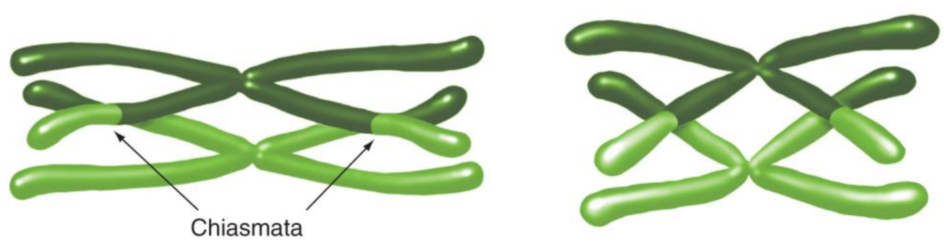
Crossing over exchanges genetic information.



(a) Leptotene: Threadlike chromosomes begin to condense and thicken, becoming visible as discrete structures. Although the chromosomes have duplicated, the sister chromatids of each chromosome are not yet visible in the microscope.

(b) Zygotene: Chromosomes are clearly visible and begin pairing with homologous chromosomes along the synaptonemal complex to form a bivalent, or tetrad.

(c) Pachytene: Full synapsis of homologs. Recombination nodules appear along the synaptonemal complex.



(d) Diplotene: Bivalent appears to pull apart slightly but remains connected at crossover sites, called chiasmata.

(e) Diakinesis: Further condensation of chromatids. Nonsister chromatids that have exchanged parts by crossing-over remain closely associated at chiasmata.

Mistakes in meiosis produce defective gametes

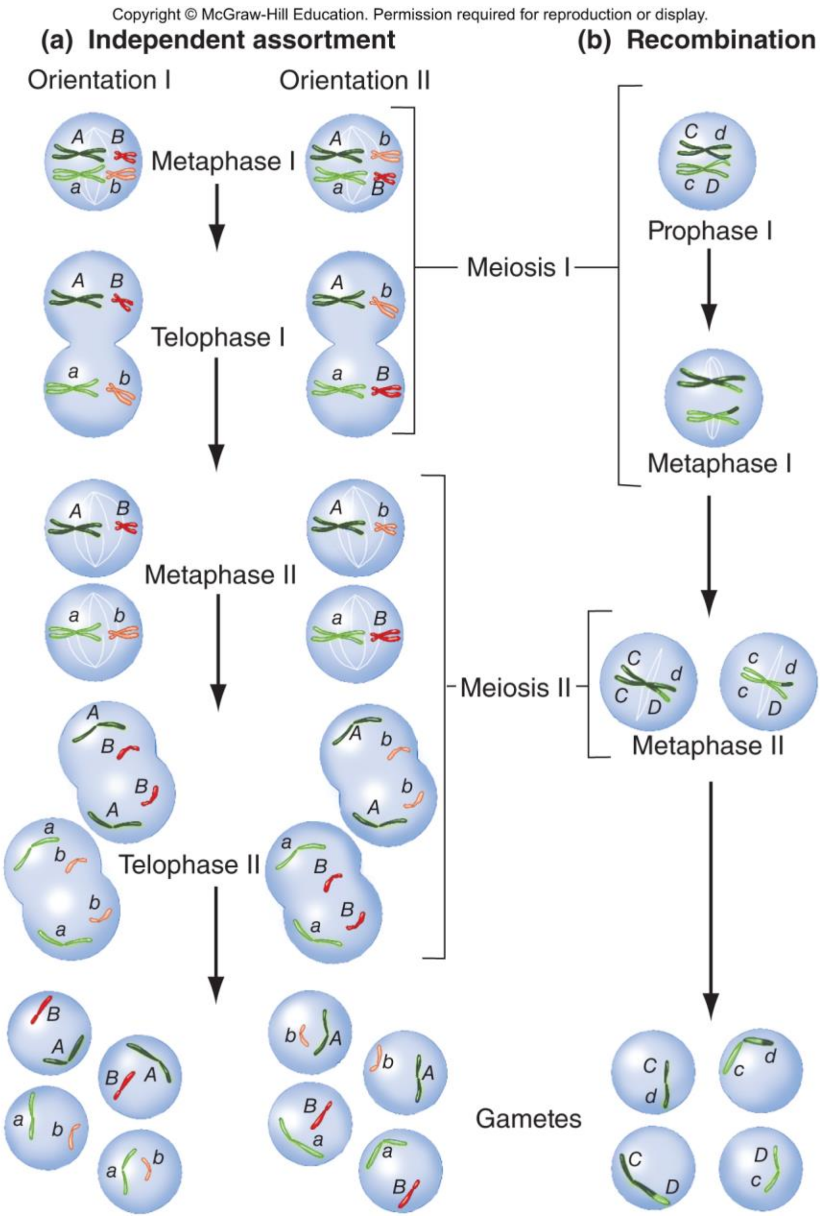
Nondisjunction – mistakes in chromosome segregation during meiosis I or II

- **May result in inviable gametes or embryos**
- **Can also result in abnormal chromosome numbers in viable individuals (e.g. trisomy 21, Down syndrome; or XXY, Klinefelter syndrome)**

Meiosis contributes to genetic diversity in two ways

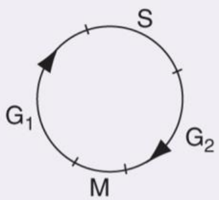
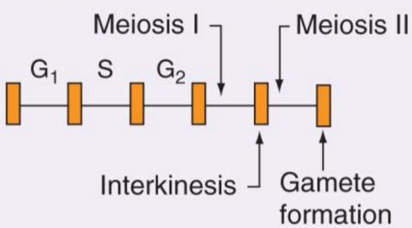
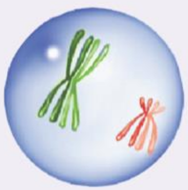
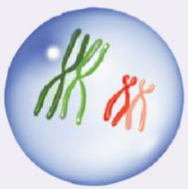
Independent assortment of nonhomologs creates different combinations of alleles

Crossing-over between homologs creates different combinations of alleles within each chromosome

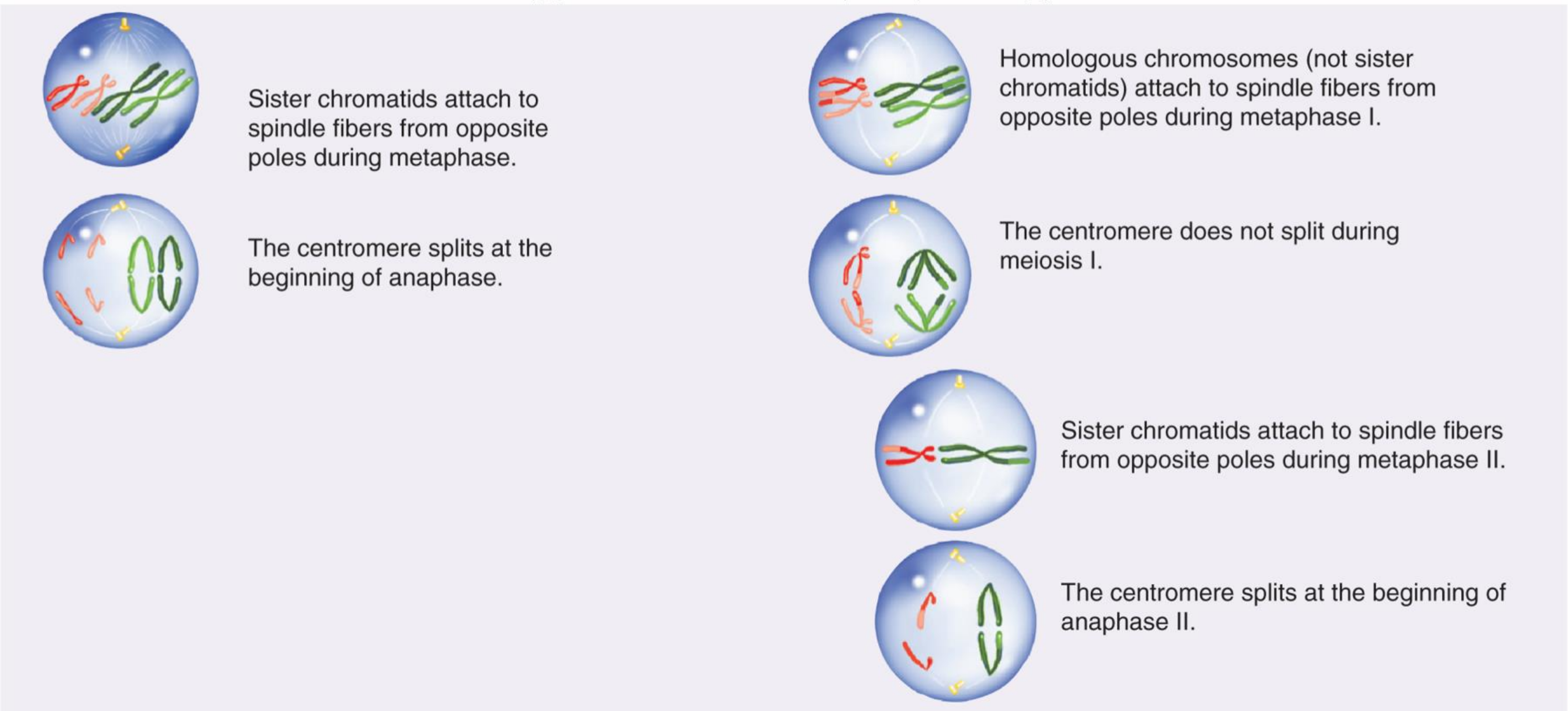


Comparison of mitosis and meiosis

TABLE 4.3 Comparing Mitosis and Meiosis

Mitosis		**Meiosis**	
Occurs in somatic cells Haploid and diploid cells can undergo mitosis One round of division		Occurs in germ cells as part of the sexual cycle Two rounds of division, meiosis I and meiosis II Only diploid cells undergo meiosis	
	Mitosis is preceded by S phase (chromosome duplication).		Chromosomes duplicate prior to meiosis I but not before meiosis II.
	Homologous chromosomes do not pair.		During prophase of meiosis I, homologous chromosomes pair (synapse) along their length.
	Genetic exchange between homologous chromosomes is very rare.		Crossing-over occurs between homologous chromosomes during prophase of meiosis I.

Comparison of mitosis and meiosis (continued)



4.5 Gametogenesis

Learning Objectives:

- **Compare the processes of oogenesis and spermatogenesis in humans.**
- **Distinguish between the sex chromosome complements of human female and male germ-line cells a different stages of gametogenesis.**

Gametogenesis in sexually reproducing animals

Germ line – specialized diploid cells set aside during embryogenesis

Gametogenesis – the formation of gametes

- **Involves meiosis as well as specialized events before and after meiosis**
- **Different types of animals have variations on general aspects of this process**
- **In humans, oogenesis produces one ovum from each primary oocyte**
- **In humans, spermatogenesis produces four sperm from each primary spermatocyte**

Oogenesis in humans

Oogonia – diploid germ cells in ovaries of female embryos

- **Divide by mitosis and enter meiosis I to become primary oocytes**
- **Primary oocytes arrest in diplotene stage of meiosis I until after birth**

At puberty, one primary oocyte per month resumes meiosis at ovulation

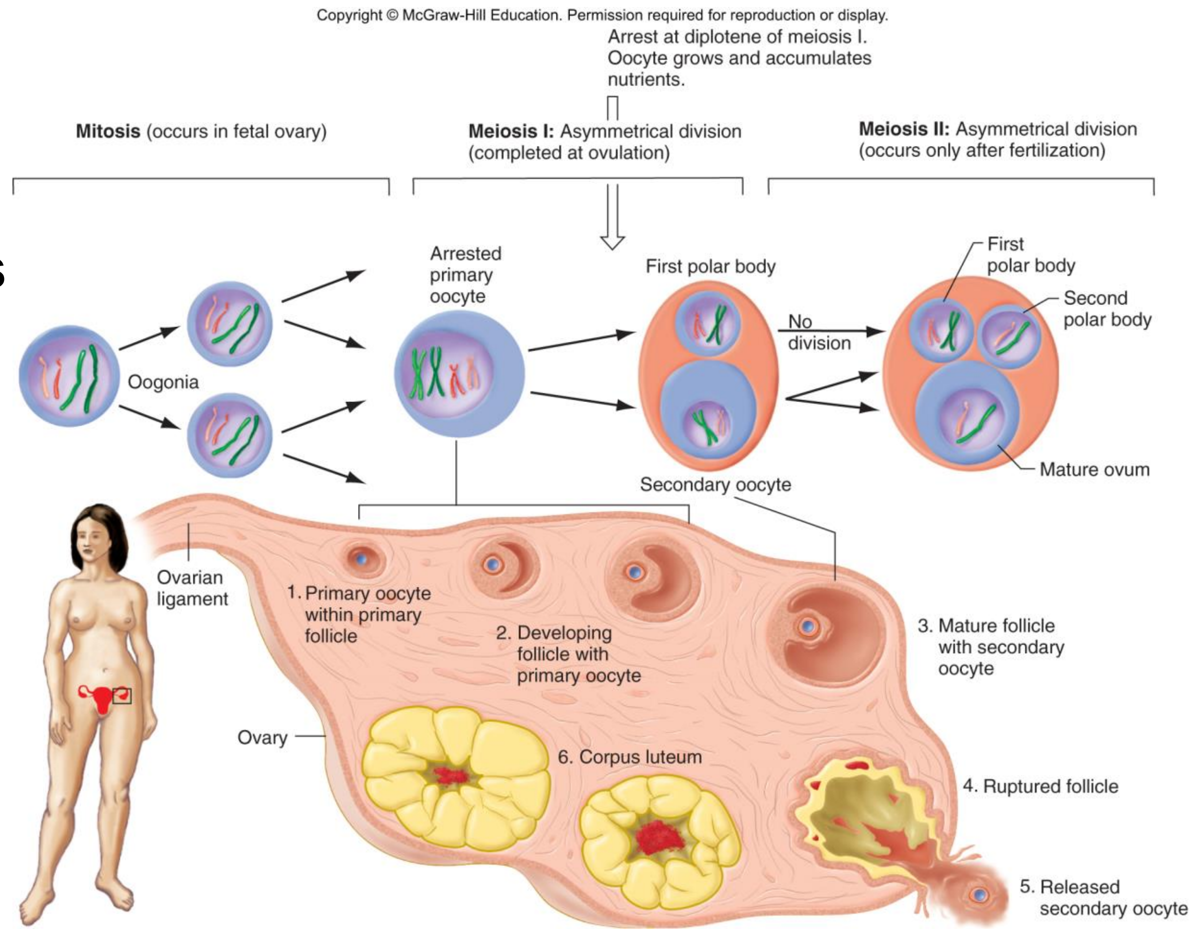
- **completion of meiosis I produces a secondary oocyte and first polar body**
- **Secondary oocyte arrests in metaphase of meiosis II**

At fertilization, meiosis II is completed and produces a mature ovum and second polar body

In humans, egg formation begins in the fetal ovaries and arrests during prophase of meiosis I

One ovum and 2 – 3 nonfunctional polar bodies produced by asymmetrical meiosis

Long meiotic arrest may contribute to chromosome segregation errors (e.g. trisomies)



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Spermatogenesis in humans

Spermatogonia – diploid germ cells found only in testis, divide by mitosis for form primary spermatocytes

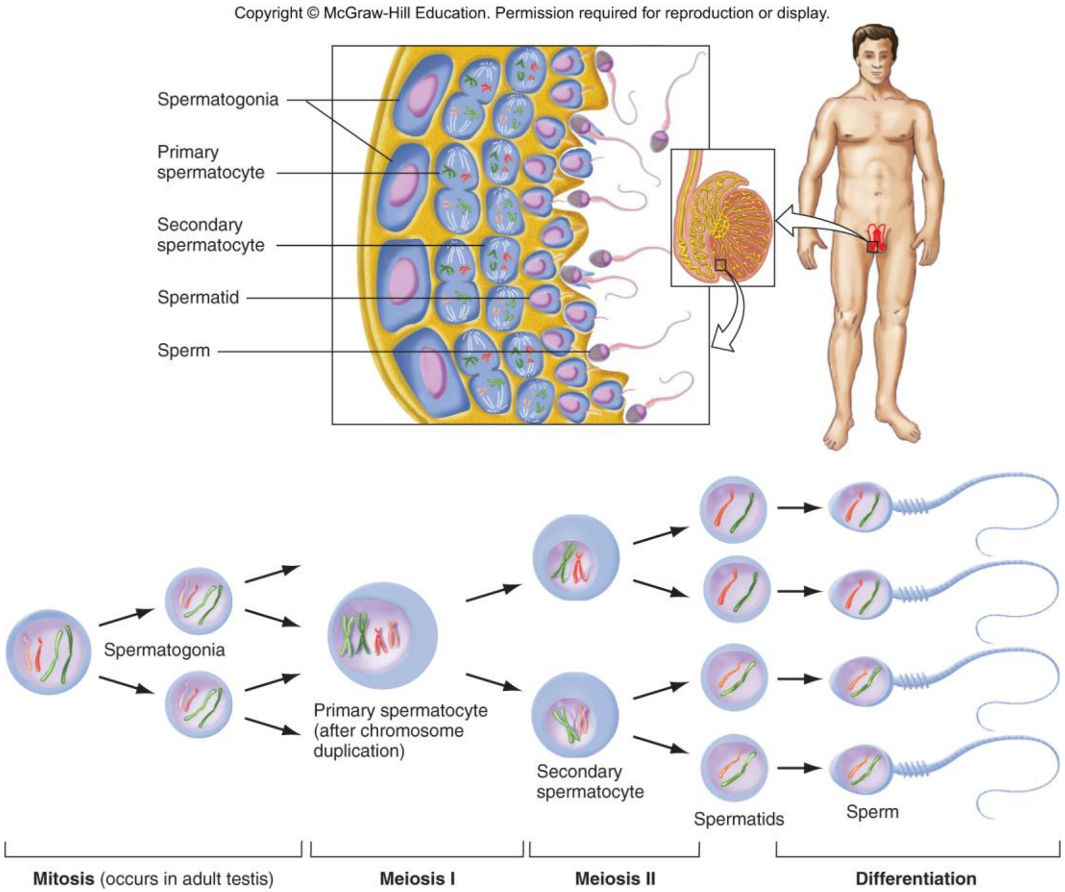
After puberty, maturation begins to form sperm

- primary spermatocytes undergo symmetrical division at meiosis I to produce two secondary spermatoctyes
- Secondary spermatocytes undergo symmetrical division at meiosis II to produce two spermatids
- Spermatids mature to become sperm
- Equal numbers of X and Y sperm are produced

Human sperm form continuously in the testes after puberty

Four haploid sperm produced by symmetrical meiosis of each spermatocyte

Mitosis and meiosis occur throughout adult life



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4.6 Validation of the chromosome theory

Learning Objectives:

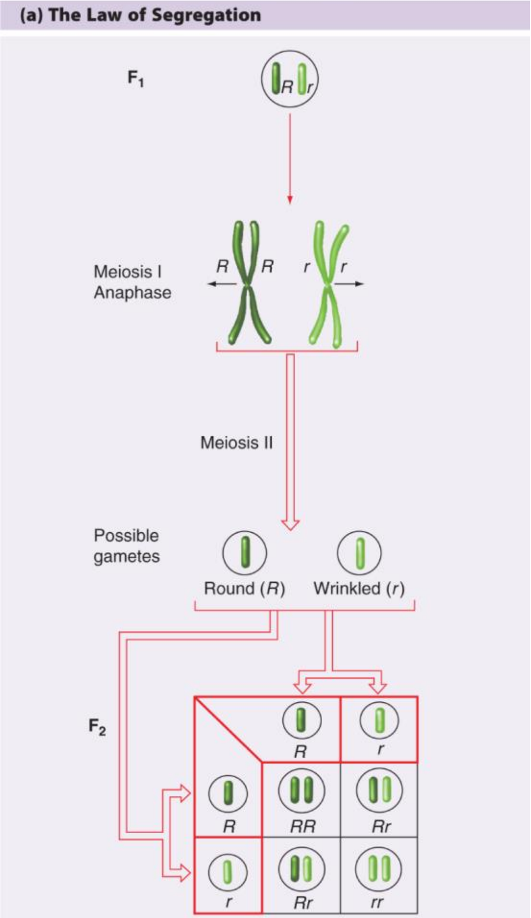
- **Describe key events of meiosis that explain Mendel's laws.**
- **Infer from the results of crosses whether or not a trait is sex-linked.**
- **Predict phenotypes associated with nondisjunction of sex chromosomes.**

The chromosome theory of inheritance

Walter Sutton – 1903, chromosomes carry Mendel's units of heredity

- **Two copies of each kind of chromosome**
- **Chromosome complement is unchanged during transmission to progeny**
- **Homologous chromosomes separate to different gametes**
- **Maternal and paternal chromosomes move to opposite poles**
- **Fertilization of eggs by random encounter with sperm**
- **In all cells derived from fertilized egg, half of chromosomes are maternal and half are paternal**

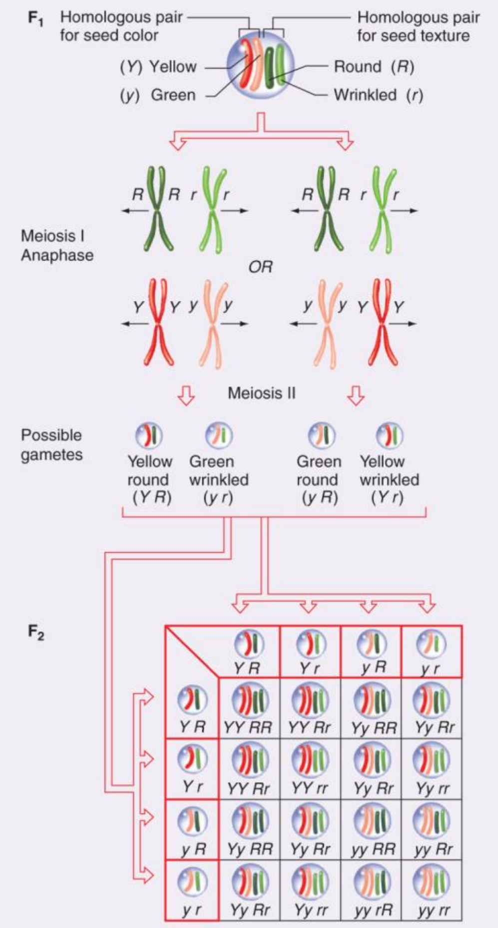
How the chromosome theory of inheritance explains Mendel's law of segregation



In an F_1 hybrid plant, the allele for round-seeded peas (*R*) is found on one chromosome, and the allele for wrinkled peas (*r*) is on the homologous chromosome. The pairing between the two homologous chromosomes during prophase through metaphase of meiosis I makes sure that the homologs will separate to opposite spindle poles during anaphase I. At the end of meiosis II, two types of gametes have been produced: half have *R*, and half have *r*, but no gametes have both alleles. Thus, the separation of homologous chromosomes at meiosis I corresponds to the segregation of alleles. As the Punnett square shows, fertilization of 50% *R* and 50% *r* eggs with the same proportion of *R* and *r* pollen leads to Mendel's 3:1 ratio in the F_2 generation.

How the chromosome theory of inheritance explains Mendel's law of independent assortment

(b) The Law of Independent Assortment



One pair of homologous chromosomes carries the gene for seed texture (alleles *R* and *r*). A second pair of homologous chromosomes carries the gene for seed color (alleles *Y* and *y*). Each homologous pair aligns at random at the metaphase plate during meiosis I, independently of the other homologous pair. Thus, two equally likely configurations are possible for the migration of any two chromosome pairs toward the poles during anaphase I. As a result, a dihybrid individual will generate four equally likely types of gametes with regard to the two traits in question. The Punnett square affirms that independent assortment of traits carried by nonhomologous chromosomes produces Mendel's 9:3:3:1 ratio.

Validation of the chromosome theory

Prior to 1910, the chromosome theory of inheritance was supported by two circumstantial lines of evidence

1. Sex determination associates with inheritance of particular chromosomes.

2. Events in mitosis, meiosis, and gametogenesis ensure constant numbers of chromosomes in somatic cells.

This theory confirmed and validated by:

1. Inheritance of genes and chromosomes correspond in every detail

2. Transmission of particular chromosome coincides with transmission of traits other than for sex determination

Nomenclature for *Drosophila* genetics

Gene symbol identified by abnormal phenotype

Wild-type allele denoted with superscript +

Recessive mutant allele denoted with lowercase

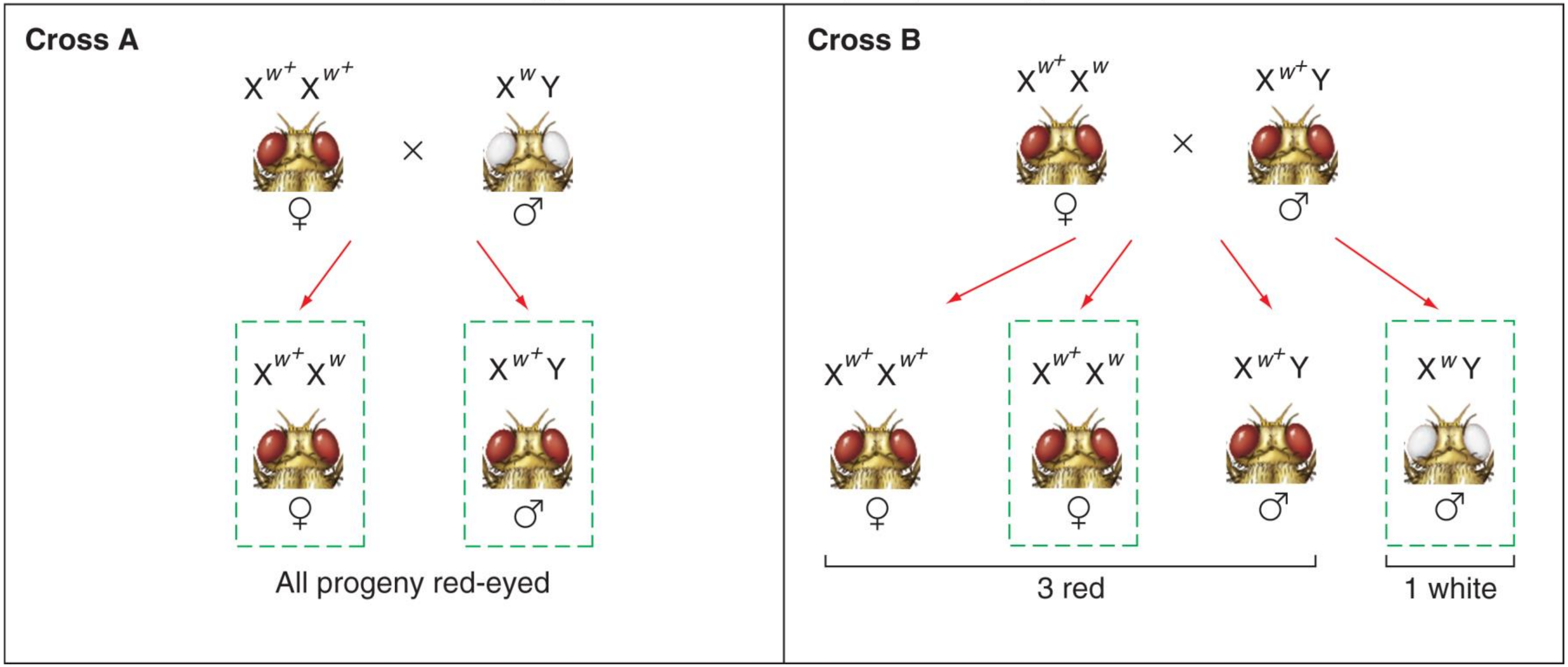
- **e.g. gene symbol for *white* gene is *w***
- **wild-type allele (w^+) specifies brick-red eyes**
- **mutant allele (*w*) specifies white eyes**

Dominant mutant allele denoted with upper case

- **e.g. gene symbol for *Bar eyes* is *Bar***
- **wild-type allele (Bar^+) specifies normal eye**
- **mutant allele (*Bar*) specifies abnormal eyes**

The *Drosophila white* gene is located on the X chromosome

T. H. Morgan (1910) discovered a white-eyed *Drosophila* mutant and did a series of crosses



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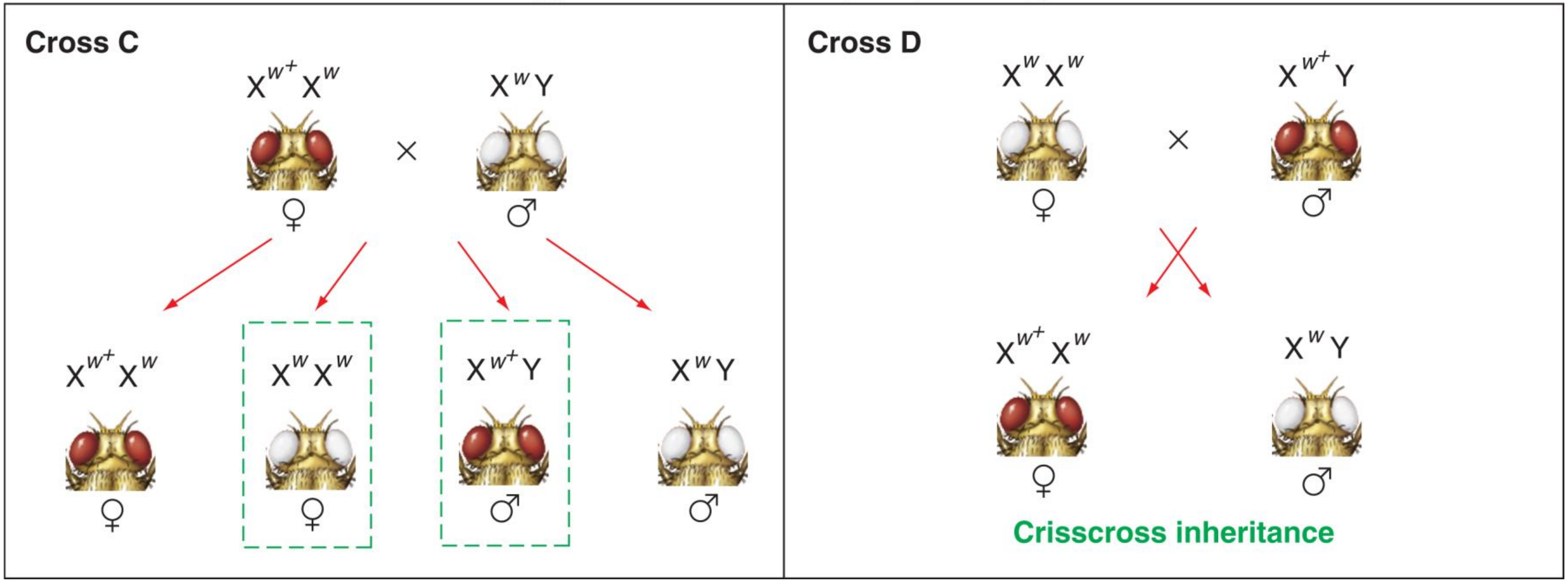
Crisscross inheritance occurs with X-linked recessive traits

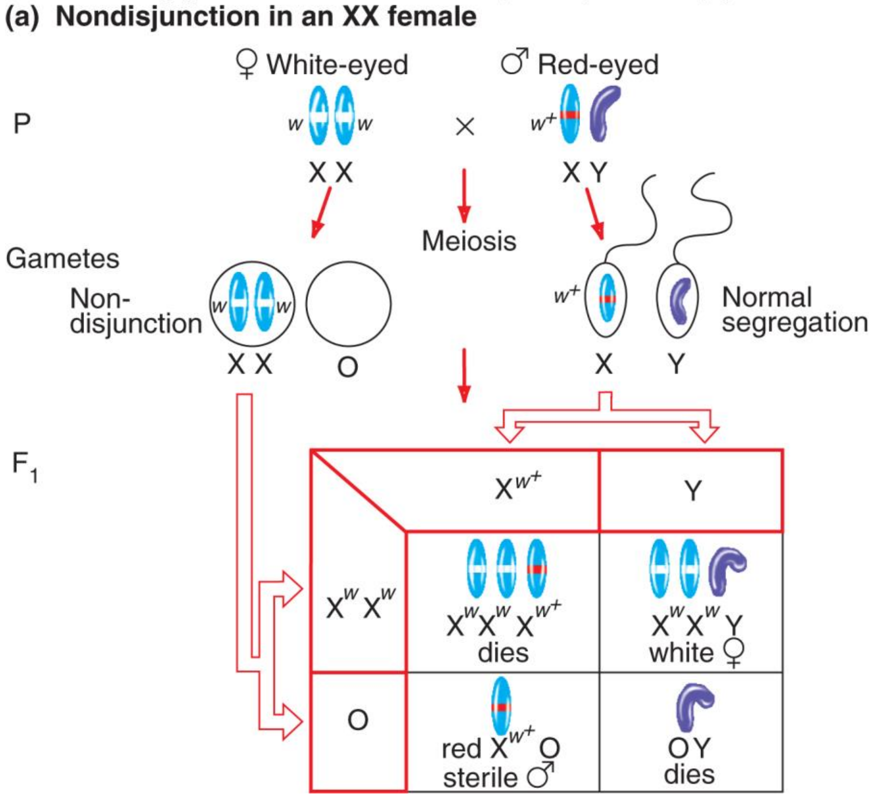
See cross D – daughters inherit the phenotype of their fathers, sons inherit the phenotype of their mothers



Rare mistakes in meiosis helped confirm the chromosome theory

C. Bridges found 1/2000 male progeny of *white* females have red eyes

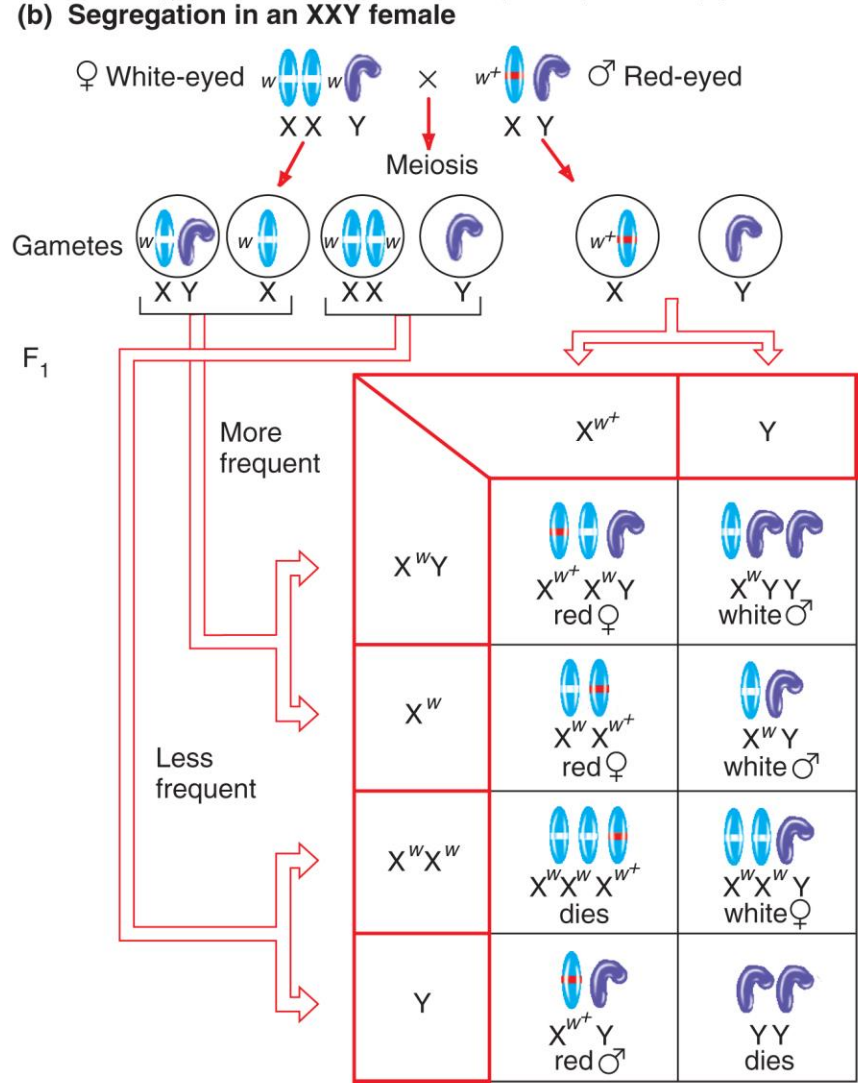
Hypothesized that red-eyed males arise from mistakes in chromosome segregation (nondisjunction) during meiosis in white-eyed females



Rare mistakes in meiosis helped confirm the chromosome theory (cont)

Chromosome segregation in an XXY female

The three sex chromosomes pair and segregate in two ways, producing progeny with unusual sex chromosome complements



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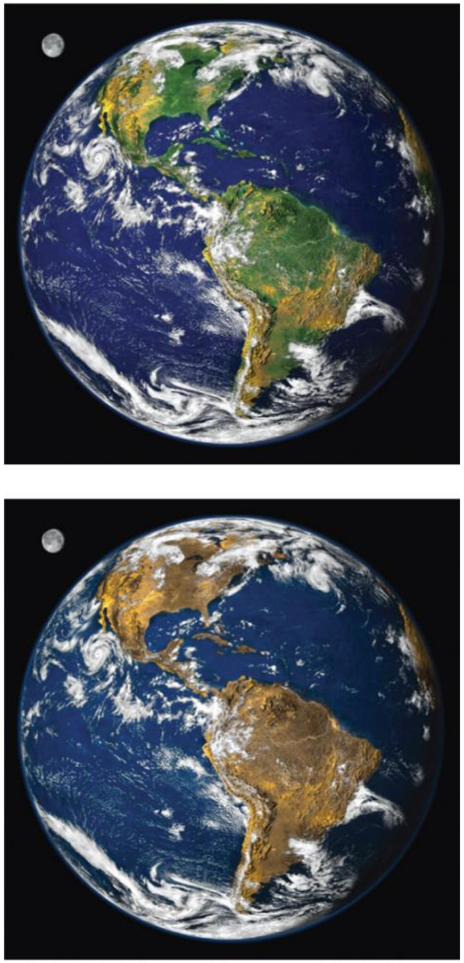
Example of an X-linked trait in humans

(Top) View of the world to a person with normal color vision

(Bottom) View of the world to a person with red-green colorblindness

E. B. Wilson – 1911, assigned gene for this trait to the X chromosome



(both): Color defi cit simulation courtesy of Vischeck (www.vischeck.com). Source image courtesy of NASA

Pedigree patterns suggest sex-linked inheritance

X-linked recessive

- Mutation never passes from father to son
- Daughters of affected males are carriers. ½ of sons of carriers will inherit the trait.

X-linked dominant

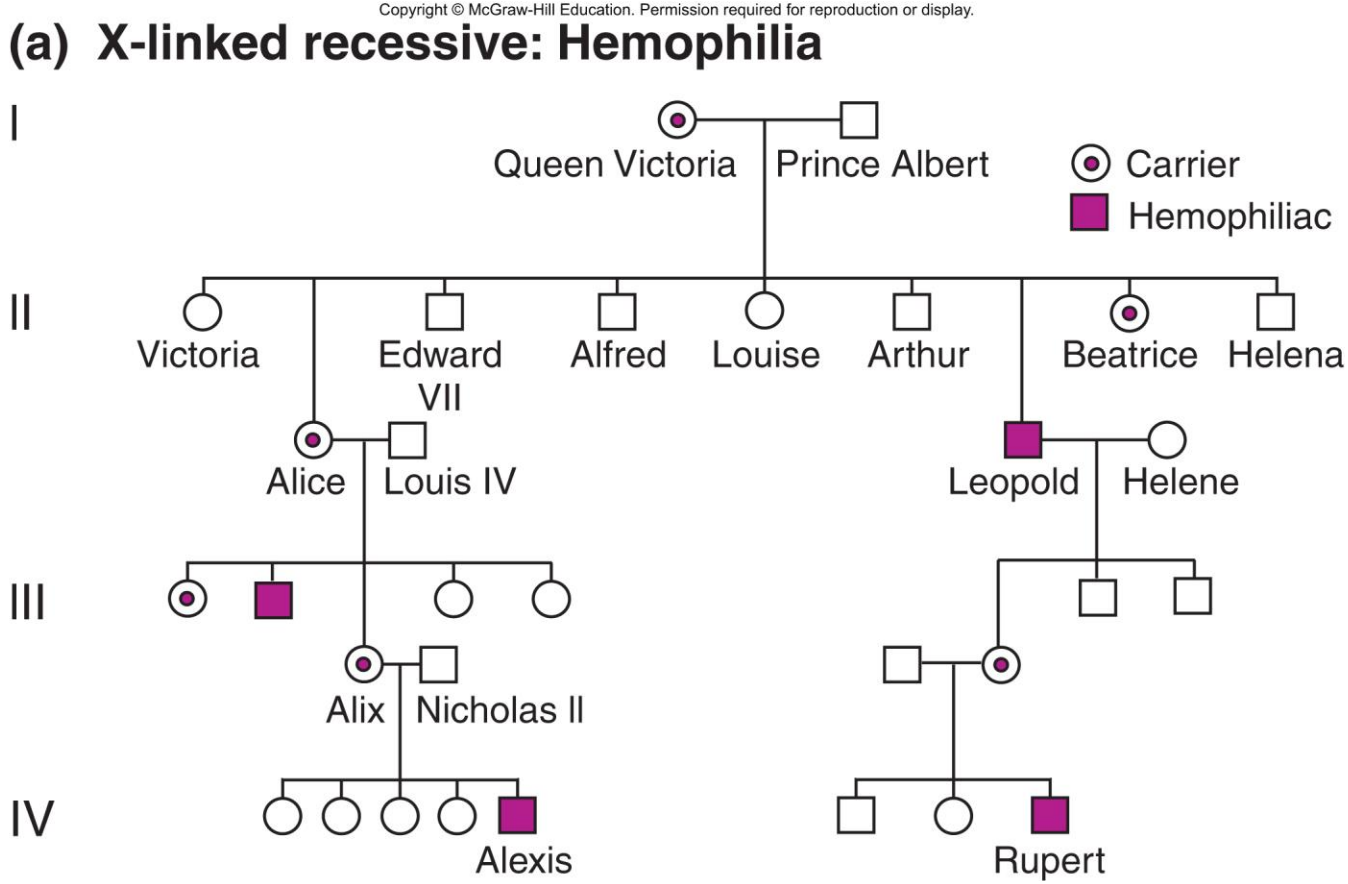
- Trait seen in every generation
- Affected male will produce affected daughters and unaffected sons.

Y-linked

- Only in males

See Table 4.5 for more patterns

An example of a pedigree for an X-linked recessive trait: Hemophilia



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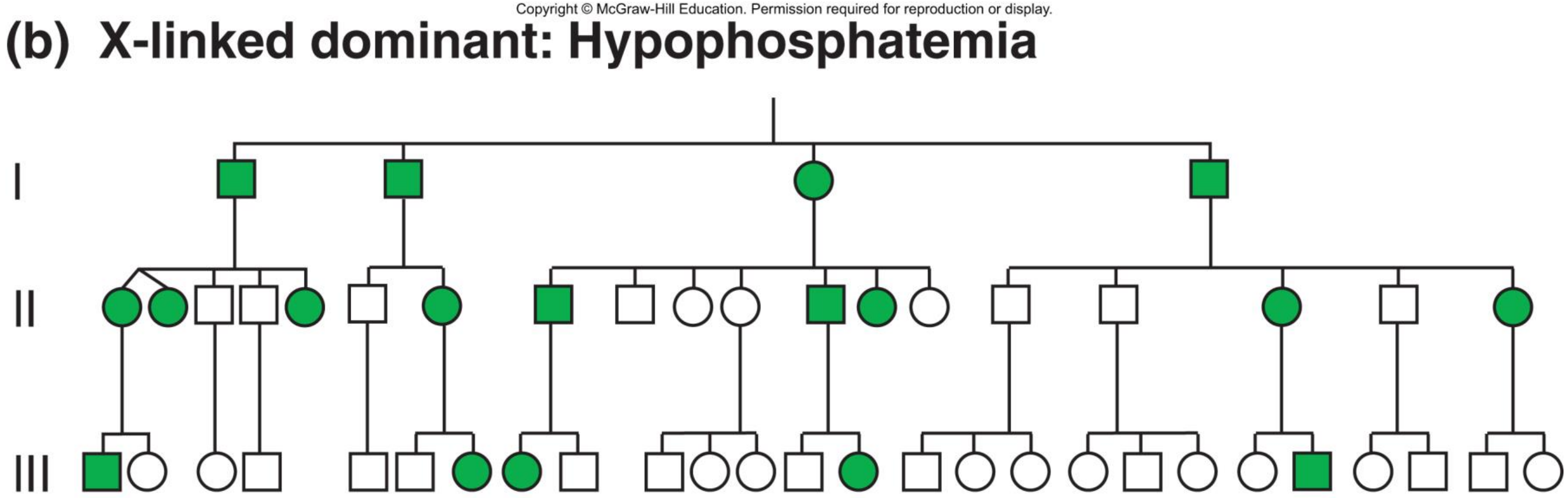
An example of a pedigree for an X-linked dominant trait: Hypophosphatemia

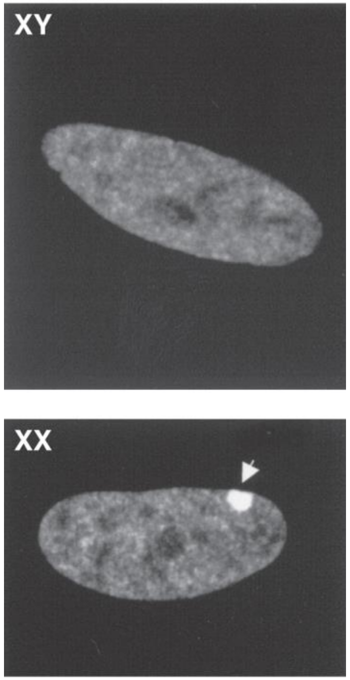


Dosage compensation

Human females have two X chromosomes and males have only one.

In female cells, one X chromosome is inactivated.

The inactive X chromosome is condensed into a Barr body.



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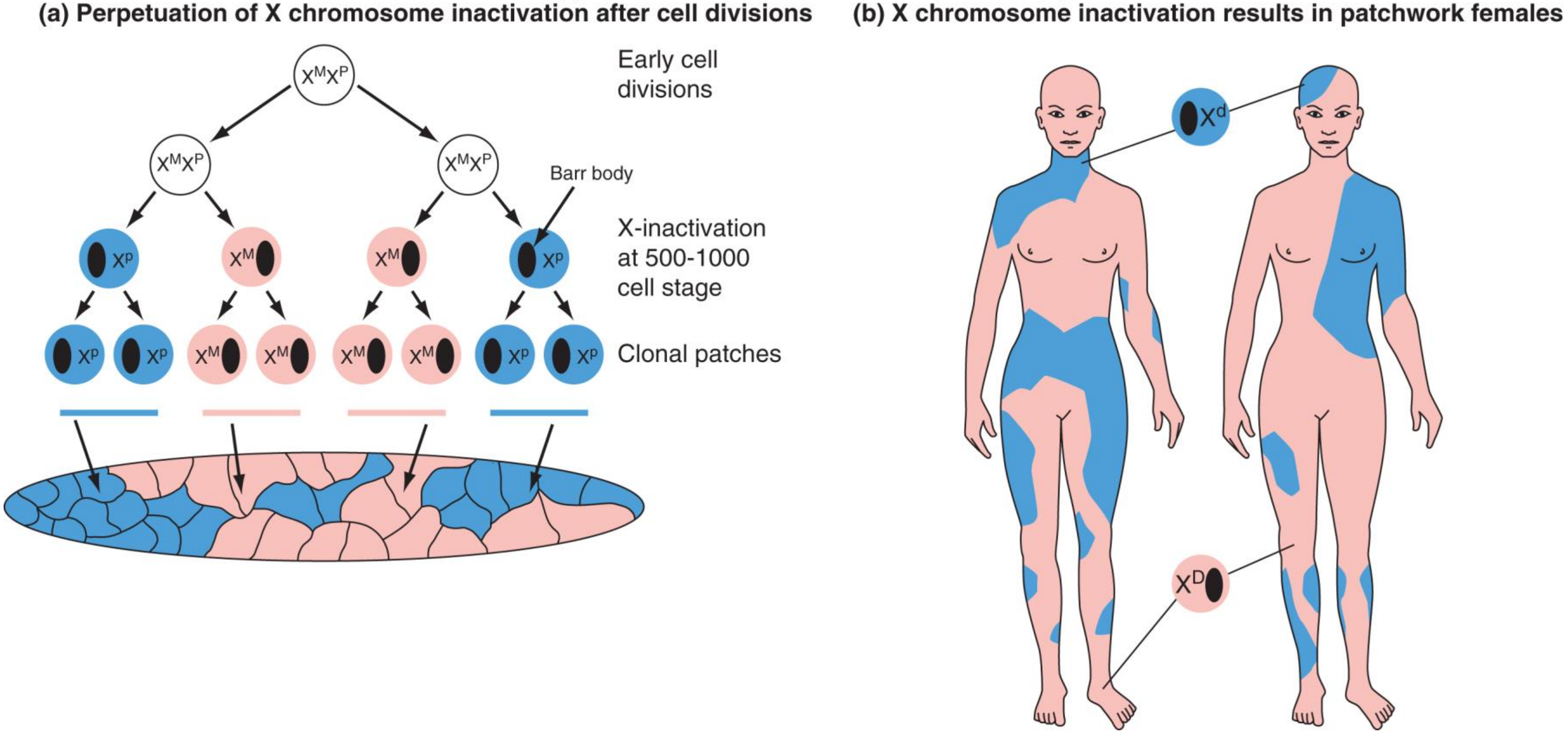
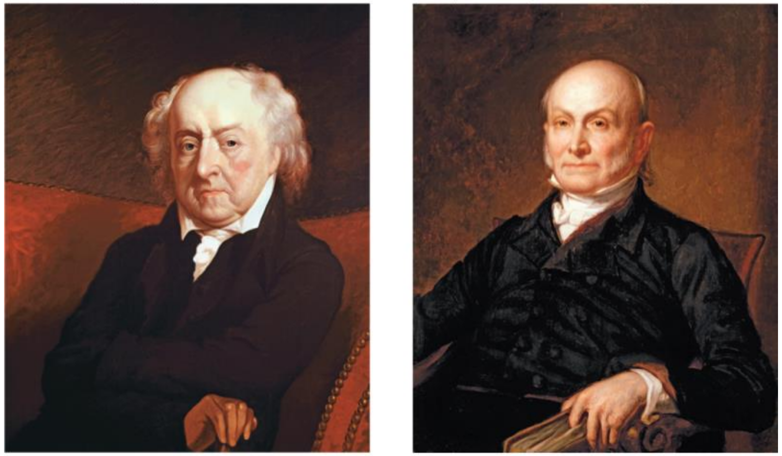
Human females are a patchwork for X-linked gene expression

In the early female embryo, each cell independently inactivates one X chromosome.



Autosomal genes and sexual dimorphism

Sex-influenced traits



(a) (b)

- **Appear in both sexes but hormonal differences may cause difference**

  e.g. male pattern baldness (right, in John Adams and John Quincy Adams)

Sex-limited traits

- **Affect a structure or process found in only one sex**
  - **e.g. Drosophila stuck mutant males can't separate from female after mating**